Chapter 6

Properties

This chapter aims at providing ontological foundations for some of the most basic conceptual modeling constructs, namely *types*, *attributes*, *data types*, and *associations*.

Types (e.g., *Person* or *Car*), attributes (e.g., *being colored*, or *being happy*), and associations (e.g., *being married to*, *being enrolled at*) are all considered sorts of *universals*, i.e., predicative terms that can possibly be applied to a multitude of individuals. In chapter 4, we have focused our discussion on types. Universals such as attributes and associations are referred here by the general name *Property*. This chapter is therefore mostly about properties.

The notion of universal underlies the most basic and widespread constructs in conceptual modeling. Therefore, before we can provide ontological foundations for these constructs, we start here by investigating the nature of universals from a philosophical point of view. We start the chapter on section 6.1 by discussing the so-called *problem of universals* in philosophy. In particular, we analyze and criticize a specific theory of universals that underlies most current conceptual modeling languages, including the *semantic web languages*. In virtue of these criticisms, we justify the choice of some theoretical entities that are incorporated in our foundational ontology, which is fully elaborated in section 6.2.

In section 6.3, we apply the proposed ontological categories to interpret the modeling concepts of types, attributes, datatypes, and associations, and to provide methodological guidelines for their use in conceptual modeling.

In section 6.4, we present a more general discussion on related work in the conceptual modeling literature. Section 6.5 closes the chapter with some final considerations.

6.1 The Problem of Universals

In chapter 4 of this thesis we have discussed different types of universals, and how they could relate to each other. In particular, we saw that some universals, such as Man or Dog provide a principle of identity to the objects they classify. In contrast, other universals, such as Red, Square or Hard, do not provide such a principle. Strawson (1959) names these two types of classifiers *sortal* and *characterizing universals*, respectively.

Before proceeding, there is an important notion that should be defined, namely the distinction between *determinables* and *determinates* (Johnson, 1921). Universals can have different degrees of determinateness. Determinates can be understood as restrictions of determinables, providing a higher degree of specificity. For instance, *being colored* is a determinable and *being red* is a determinate for it. Since these are relative notions, *being red* can also be considered a determinable for *being scarlet* (one of its determinates). (Funkhouser, 2004) uses the terms *super-determinable* and *super-determinate* for the universals that are the root and the leaves of a specialization chain, respectively.

Now, take for example a sortal like Person. For every individual person, say John, there are many characterizing universals that apply to John, in virtue of John being of that specific kind. For example, John can be *1.80 meters tall*, *80 kilograms of weight*, be *29 years old*, and so forth. Notice that properties such as *being colored*, *being red*, *being-1.80-meters-tall* or *being married* are indeed sorts of universals, since they can be multiple instantiated in different individuals (i.e., they are repeatable). This gives us the idea that *sortal universals form clusters of characterizing ones*. Actually, most of the categories in the profile proposed in chapter 4 are typically used to model clusters of characterizing universals, with the exception of *mixins*. Sometimes characterizing universals are represented as mixins for the purpose of improving the structure of the resulting models, but more typically they are represented in conceptual models as attributes (e.g, color), in the case of super-determinables, or as attribute values (e.g, red), in the case of determinates. In fact, (Guarino & Welty, 2000) use the term *Attribution* for mixins, highlighting the correlation between the two representations of characterizing universals.

The issue of universal properties has been a topic of great interest (and controversy) throughout the entire history of western philosophy, from Plato and Aristotle to early analytic philosophers such as Russel and Wittgenstein, and contemporary philosophers such as Bunge (1977), Boyd (1991), Armstrong (1989, 1997) and Thomasson (2004). This topic is known in philosophy as *"the problem of universals"* (Loux, 2001) and can be summarized in the following manner. We know that proper names (e.g., Noam Chomsky or Spot) refer to individual entities, but what do general terms (or universal properties) refer to (if anything at all)? We classify objects as being of the same type (e.g., person) and use the same predicate or general term (e.g., red) to different objects. What exactly *is the same* in different objects that justify their belonging to the same category? In the sequel, we briefly present different approaches in answering these questions that amount to different philosophical theories of universals.

6.1.1 Lightweight Ontologies and the *Class* Theory of Universals

A first group of theories is constituted by what we name here *Class theories of universals* (henceforth, *C-theories*). In a C-theory, the meaning of a type is associated with that of class in the mathematical sense, i.e., roughly a set. Hence, if *a* and *b* are of the same type, it is because they belong to the same set X, where X is the ontological interpretation of that type.

In the sequel we discuss C-theories in some detail, mainly because there are many languages in conceptual modeling that commit to this view of universals. Examples include OWL (Bechhofer et al., 2004), LINGO (Falbo & Menezes & Rocha, 1998; Guizzardi & Falbo & Pereira Filho, 2002), CCT (Dijkman & Ferreira Pires & Joosten, 2001); Z (Spivey, 1988), ER (Chen, 1976), among others. The choice for an underlying class ontology in these languages is sometimes motivated by mathematical simplicity and convenience, sometimes related to practical performance trade-offs. For example, LINGO was designed with the specific objective of achieving a positive trade-off between expression power of the language and the ability to facilitate bridging the gap between the conceptual and implementation levels (a preoccupation that also seem to be present in Peter Chen's original proposal for ER diagrams). OWL, on the other hand, has been designed to explore the relation between a minimum expressiveness and computational tractability in the reasoning procedures. Although these choices can be justifiable in *lightweight* ontology modeling languages for practical reasons, there are many philosophical problems that would make them unsuitable as *foundational*[43] conceptual modeling languages.

To start with, a major problem with the simplest form of a C-theory is the difficulty in correlating classes and types. If C-theory is to be true, there should be a one-to-one correlation between the two. However, this correlation is not to be found. Take the set-theoretical semantics given to some of the modeling languages aforementioned (e.g., CCT):

– a type is interpreted as a set;

[43] See the distinction between lightweight and foundational ontologies in section 3.3.3 of this thesis.

– the subtyping relation is interpreted as the subset relation;
– the instantiation relation is interpreted as the membership relation.

In this case, the following problems are immediately found:

a1. If the subtyping relation between types A and B is interpreted as the subset relation between their extensions, then any type C with an empty extension (i.e., that has no instances) can be considered a subtype of any other type (the empty set is a subset of any other set). Moreover, any two types C and D with empty extensions (e.g., unicorn and centaur) become identical;

a2. Starting with an individual, for instance the person John, one can construct an infinite series of sets, all which have exactly the same content (Sowa, 2000): the singleton set {John}; the set {{John}}; the set {{{John}}}, {{{{John}}}}… and so forth. Although, in set-theoretic terms, these are all different entities, they correspond to the same entity in reality. In fact, this type of *ontological extravagance* of set-theory was one of the motivations for the proposal of formal mereologies (see section 5.1), since in extensional mereology there is no distinction between an entity A and the sum of its parts. This is captured in the following statement by (Goodman, 1956), one of the fathers of mereology: *"No distinction of individuals without distinction of content."*

a3. As discussed in chapter 4, the categorization mechanism employed by human cognition have the purpose of supporting learning, improving inductive knowledge, making memory and language possible, etc. Now, take the set formed by the last thought I had on last Christmas, the number two, the apple over my table and the last two minutes of a football game. This is a perfectly acceptable set (from a mathematical point of view), but one which has no place in a cognitive model of reality.

An attempt to solve some of these problems is made by a refinement of the pure C-theory, named the *Natural Class theory*. This theory was first proposed by philosopher Anthony Quinton (1957), who proposes a fundamental distinction in the world between *natural* and *unnatural* classes. According to Quinton, it is a feature of the world that things fall into natural classes, and this primitive notion of belonging to a natural class cannot be further analyzed. Thus, particulars, by virtue of falling *naturally* into a class, can be said to be of this or that type. In other words, the meaning of a type (e.g., horse) is the natural class of things that are instances of that type, i.e., the set of individuals that are members of the *natural class of Horses*. By recognizing that only certain sorts of classes are related to a type, the Natural Class theory can address the criticisms (a2) and (a3) above. In particular, the set mentioned in (a3) would lack naturalness and therefore would not be associated to a type.

To meet the criticisms stated in (a1), one could just go a little bit further and postulate that the empty class is not natural. However, underlying the criticisms in (a1) there is a more general problem, due to the extensional principle of identity of sets. This principle of identity, know as the *extensionality principle*, states that two sets are identical iff they have the same members. According to this principle then, the following consequences are implied:

b1. If a class is taken to be the ontological interpretation of a type, then the identity of classes become the identity of corresponding types. According to the extensionality principle, if a class gains or looses a member it becomes a different class. Take for example the type Electron. If the meaning of this type is the class of existing electrons, then addition or deletion of any electron would change the meaning of what is to be an electron. This is taken by many philosophers (and we agree) to be an absurd consequence;

b2. Any two classes that happen to be co-extensional must be considered identical. The case of empty classes mentioned previously in (a1) is just an extreme case of this problem. Take for example the types *human* and *featherless biped*, or *liquid* and *viscous.* These types have completely different intentions and can have different associated principles of identity (see chapter 4). The fact that two types happen to coincide in their extensions should not be sufficient for equating them;

b3. Suppose a subtyping relation between the types Organization and Group of People in the loose sense that “every organization (e.g., musical group, football team, bird-watchers association) is a group of people”. If subtyping is interpreted as the subset relation, then this is necessarily true in this case. However, as discussed in chapter 4, it can be the case that these universals supply incompatible principles of identity for their instances. For instance, if a football team exchanges one of its members it is still the same organization but it becomes a different group of people. As a consequence, despite of apparently sharing the same extension these two types cannot be related via subtyping, since they carry incompatible principles of identity.

Consequently, a minimum requirement of a set-theoretic conceptual modeling language would be the explicit consideration of *possible worlds*. In other words, the extension of natural class should consider individuals in all possible worlds[44]. This way, the theory can exclude all cases where the extensions of classes coincide only contingently. To cite one last example, it is a fact in our world that the classes of cordates (individuals that have a heart) and of renates (individuals that have a kidney) coincide. Now, suppose that this is just an evolutionary accident, in the sense that there is no fundamental *direct functional dependence* (see chapter 7) between hearts and kidneys. Therefore, one can conclude that there are possible worlds in which the extensions of cordate and renate do not coincide and, since the two classes do not coincide necessarily, they cannot be said to be the same. Moreover, by considering Natural classes as the extension of types in every possible world, the theory becomes more refined, and can choose to deem as unnatural empty classes, only those that are necessarily empty. Finally, it can also escape criticism (b1) regarding the dependence that the meaning of type has on its members in its extensional counterpart. Notice that there is support in the cognitive psychology literature for taking the meaning of a general term to refer to particulars in counterfactual situations. For example, in the second of the psychological claims defended by John Mcnamara in (Mcnamara, 1994) it is proposed that "the word DOG refers to all dogs that ever existed, that ever will be, or that could ever have been. Not to the dogs that currently exist."

Extending the notion of natural class by considering possible words would solve some of the problems related to the ontological semantics of languages such as EER, Z, CCT and LINGO, when considered as conceptual modelling languages. However, there is still another philosophical disadvantage of the natural class approach. It is related to the so-called *direction of explanation* (Amstrong, 1989). Consider the following question: Is the sort of thing something is – say a horse - because it is a member of the class of Horses? Or it is rather a member of the class of Horses because it is a horse? In conformance with Armstrong, we claim the latter is the correct direction of explanation.

This becomes more evident when considering the *causal action* of things. Things act causally by virtue of the *properties* they have, not because they belong to certain classes. For instance, the fire makes the water boil by virtue of its temperature and the object depresses the scale by virtue of its mass. It is the individual *four kilograms object* that acts on the scale in a way that is completely independent of the other four kilograms objects that exist (or may have existed). In other words, it is not the case that the whole class of four kilograms object is relevant for the causal action of an object on another, but the presence of certain property of the object. Finally, from an epistemological point of view, acknowledging the priority of properties over

[44] See our function *ext* (as opposed to ext_w) in section 4.1.

classes seems to afford a more intuitive explanation for our ability to classify things. Suppose that we face a new particular of a familiar type and that we have no difficulty in correctly classifying it as being of that type. If this is the case, what happened is that the object has acted on us by virtue of certain properties. To use a simplistic example, we correctly apply the term Red to an object because we recognize the color red in it, not because we recognize it as a member of a (possible infinite) class of objects.

We therefore propose an account of universals that recognize the precedence of properties over classes, and the precedence of intention over extension. Therefore, in the world view proposed here objects are endowed with a number of properties that make them what they are.

There is a final problem with C-theories that is worth mentioning, since it is related to a recurrent practice in some modeling languages, in particular, in the so-called *Semantic Web Languages* (Bechhofer et al., 2004; McGuinness et al., 2002a,b). This problem becomes even more evident in a particular C-theory that we name here the *predicate theory.* Let us first briefly explain the predicate theory.

The predicate theory of universals belongs to the group of the so-called *nominalist* theories. Nominalism (from the latin word *Nomem* for *name*) is a project of explaning the unity of the tokens falling under a certain type by some linguistic device (Armstrong, 1989). According to it, two things are of the same type iff the same predicate applies to them. To put it differently, the only thing that is universal among, for example, two red objects x and y is the fact that we apply the linguistic predicate "red" to both of them. This is a serious problem from a philosophical point of view because it commits to the idea that there are no real universal properties shared by individuals outside the minds of cognitive subjects. Even worse, it equates the set of universal properties to the set of predicates available in a language. Other C-theories such as the natural class theories do not make this commitment. The naturalness of a class (or lack of it) is taken to be a feature of reality and, thus, independent of language.

However, there is a problem with predicate theory that is also common to the simplest form of C-theories. There are many predicates that can be constructed in the language that do not correspond to properties in reality, in particular, this is the case for the *negative* and *disjunctive predicates* (Bunge, 1977). Take, for example, the complex predicate $P(x) =_{def} C(x) \lor M(x)$ which is true for all objects x that have either an electrical charge or a mass. Although, P is a perfectly good predicate, it is not a universal (type) in the philosophical sense. To see that, consider the following situation: let a and b be two individuals such that both $P(a)$ and $P(b)$ hold. $P(a)$ holds because a has an electrical charge and $P(b)$ holds because b has a mass. Now, from the fact that the disjunctive predicate P applies to both a and b, can one say that in any serious sense that a and b have something in common? Asides from that, as put by (Armstrong, ibid.), there is a close link between universals and causality. If a thing instantiates a certain universal then, in virtue of that, it has a certain power to act in a certain way. Moreover, different universals bestow different powers. For instance, by virtue of having a certain mass, *b* can act upon a scale. Likewise, by virtue of having a certain charge, *a* can repel certain things. Now, were P a genuine predicate of *a* (or *b*), it would add something to *a*'s (or b's) capacity to act. This is clearly not the case. A similar case can be made for opposing negative predicates. Although the predicate $Q = \neg C(x)$ is a perfectly acceptable predicate, it does not correspondent to a bonafide universal in reality, since: (i) there is nothing necessarily in common between two things to which Q applies; (ii) there are no causal powers which are bestowed by not instantiating a universal. In summary, in a foundational ontology, there should be no negative or disjunctive universals (Bunge, 1977; Armstrong, 1989; Schneider, 2002).

We can summarize the discussion in this section by considering the following: mathematical set (class) theories have been developed in advanced forms for the last hundred years. The idea that we can give an account in set-theoretical terms of what is to be of certain type is attractive to logicians and mathematically inclined computer scientists. However, as we have made clear in the course of this section, developing purely set-theoretical semantics for classifiers in conceptual modeling amounts to one of the exemplar cases in which formal semantics, for the sake of mathematical convenience, has been given an unfortunate precedence over real-world semantics.

In a foundational ontology, the account of universals must be an intensional, not an extensional one. As discussed in depth in chapter 4, it is not the case that all properties that apply to an individual should be seen as ontologically equivalent. Some universals are sortal, thus providing a principle of individuation, persistence and identity. Other universals are merely characterizing. Some properties apply contingently to their instances, others essentially. In line with several proposals in the literature (e.g, Guarino & Welty, 2000, 2002b, 2004; Welty & Guarino, 2001; Gupta, 1980; van Leeuwen, 1991), we advocate that a full account of these distinctions is a fundamental feature for any conceptual modeling theory of universals. One, which is, unfortunately, ignored in many current conceptual modeling and ontology representation languages.

6.2 Basic Ontological Categories

In this section, we start collecting some categories that have been proposed in previous chapters and, in the light of what has been discussed in the previous section we aim at constructing the backbone of our foundational ontology.

As discussed in the previous section, a bicategorial ontology only based on sets and its members (as supplied by set theory) does not constitute a suitable inventory of formal entities that can be used to model reality. The approach presented here preserves set theory as a part of the foundational ontology proposed. Thus it accepts set membership as one of the ontologically basic (formal) relations that are adopted. At the same time, however, it introduces a number of other ontologically basic entities and relations. These entities are discussed in the sequel.

6.2.1 Sets and Urelements

A fundamental distinction in this ontology is between the categories of the so-called *urelements* and *sets*. Urelements are entities that are not sets. They form an ultimate layer of entities without any set-theoretical structure in their build-up, i.e., neither the membership ($\in$) relation nor the set inclusion ($\subseteq$) relation can unfold the internal structure of urelements.

We assume the existence of both urelements and sets in the world and presuppose that both the impure sets and the pure sets[45] constructed over the urelements belong to the world. This implies, in particular, that the world is closed under all set-theoretical constructions.

Here, we do not discuss any particular version of set theory, but only employ it to formally characterize aspects of other entities in our ontology. Therefore, we use common set theoretical operations such as element membership, set inclusion and proper inclusion ($\subset$), union ($\cup$), intersection ($\cap$) and difference ($\setminus$) without providing any formalization. This is partly because many formal specifications of set theory already exist and are widely available (e.g., Heller et al., 2004), but also because this neutrality allows the ontology to be extended in that regard to suit particular needs, where one could choose for particular versions of set theory. Examples are the systems of Zermelo-Fraenkel and Neumann-Bernays-Gödel (Weisstein, 2004), but also theories of quasi-sets (Kreuse, 1992) can in principle be adopted[46]. In particular, sets (and associated notions) are assumed to play a fundamental role here in the formal

[45] Pure Sets can only contain other sets as members. Impure Sets can also contain urelements (Weisstein, 2004).

[46] In the current version of our ontology we exclude the so-called quasi-objects, i.e., individuals that have determinate countability but indeterminate identity (Lowe, 2001). The reason for that lies in the purpose of our ontology (a commonsense based one) and the associated belief that quasi-objects do not exist in the mesocopic level. However, a theory of quasi-sets would have to be incorporated in a possible extension of our theory that admits quasi-objects.

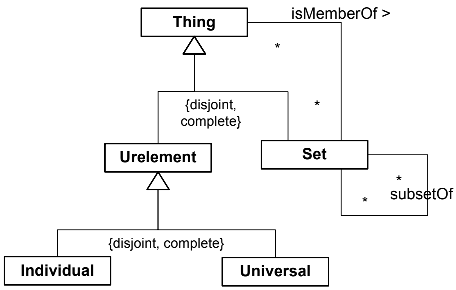
characterization of conceptual spaces, quality dimensions and quality domains (see section 6.2.5).

Urelements are classified into two main categories: *individuals* and *universals*. A urelement has to be either an individual or a universal, but not both. This can be expressed by the following axioms:

(1). $\forall x\ (Ur(x) \leftrightarrow Ind(x) \vee Univ(x))$
(2). $\neg \exists x\ (Ind(x) \wedge Univ(x))$

The categories of Set and urelements (individuals and universals) and some of their interrelations are illustrated in figure 6.1.

Figure 6-1 Fundamental distinction between sets and urelements

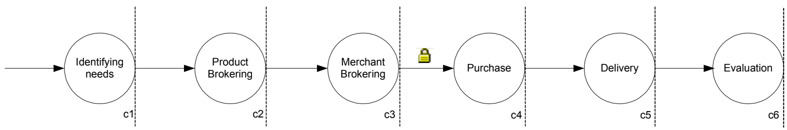


6.2.2 Individuals

Traditionally, in the philosophical literature, there is a fundamental distinction in the category of individuals between enduring and perduring entities (see, for instance, the distinction between endurants and perdurants in DOLCE (Masolo et al., 2003, or the distinction between 3D (presentials) and 4D individuals (processes) in GOL (Heller et al., 2004; Heller & Herre, 2004). Classically, the distinction between enduring and perduring individuals (henceforth named *endurants* and *perdurants*) can be understood in terms of their behavior w.r.t. time. Endurants are said to be wholly present whenever they are present, i.e., they *are in time*, in the sense that if we say that in circumstance c_1 an endurant *e* has a property P_1 and in circumstance c_2 the property P_2 (possibly incompatible with P_1), it is the very same endurant *e* that we refer to in each of these situations. Examples of endurants are a house, a person, the moon, a hole, an amount of sand. For instance, we can say that an individual John weights 80kg at c_1 but 68kg at c_2. Nonetheless, we are in these two cases referring to the same individual, namely the person John.

Perdurants are individuals composed of temporal parts, they *happen in time* in the sense that they extend in time accumulating temporal parts. Examples of perdurants are a race, a conversation, a football game, a symphony execution, a birthday party, the Second World War and a business process. Whenever a perdurant is present, it is not the case that all its temporal parts are present. For instance, if we consider the business process "buy product" at different time instants when it is present (figure 6.2), at each time instant only some of its temporal proper parts are present. As a consequence, perdurants cannot exhibit change in time in a genuine sense since none of its temporal parts retain their identity through time. Whereas for an endurant we can say that the very same individual John changes his weight from from 80 kg in c_1 to 68 kg in c_2, if we say that "buy product" has the property of being electronically secure at c_3 but non-secure in c_1, there are different proper parts of "buy product" that exhibit these properties. In a figure 6.2, "buy product" is an instance of a social process (the Consumer-Buyer Behavior model, see for instance, Almeida & Guizzardi & Pereira Filho, 2000) whereas its temporal proper-parts are actions, i.e., intentional events (atomic processes) performed by (physical or social) agents.

Figure 6-2 A "buy product" business process execution



As previously mentioned, the urelement part of our foundational ontology complements set theory (and, thus the current set theoretical approaches present in the literature) by encompassing a number of basic ontological entities. In conformance with the motivations presented in chapter 3, the ontology proposed here accounts for a descriptive commonsensical view of reality, focused on structural (as opposed to dynamic) aspects. For this reason, our foundational ontology shall be an *ontology of endurants*. Therefore, from now on, we focus our discussion on endurant individuals and endurant universals.

The core of the urelement fragment of our ontology amounts to a so-called *Four-category ontology*. The idea of an ontology centered on the four specific categories highlighted in figure 6.3 comes originally from the second chapter of Aristotle's *Categories*[47]. However, it finds support in many works in contemporary philosophical literature (Lowe, 2001; Mulligan & Simons & Smith, 1984; Smith, 1997; Neuhaus & Grenon & Smith, 2004) and, in particular, in some of the foundational ontologies developed in

[47] See, for example, the 1984 english translation (Aristotle, 1984).

computer science (Heller et al., 2004; Herre & Heller, 2004; Schneider, 2002; see also the Basic Formal Ontology in Masolo et al., 2003). The categories comprising this ontology are two pairs individuals-universals, namely substantial and substantial universals; moments and moment universals. Individuals are discussed in the sequel. Universals are discussed in section 6.2.5.

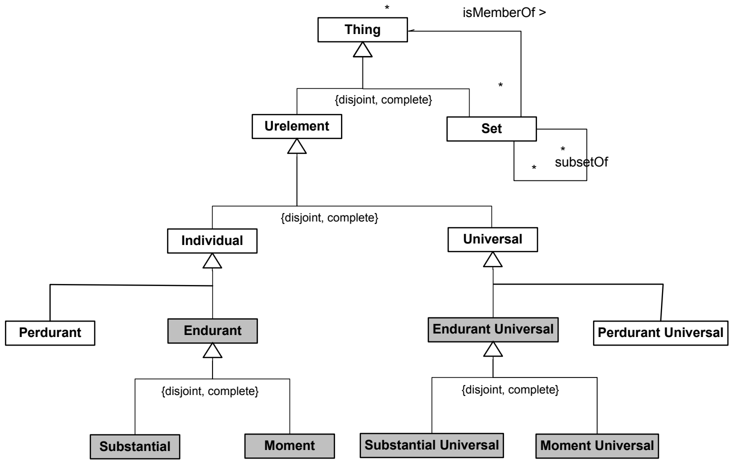


Figure 6-3 The urelement fragment on the proposed foundational ontology centered in a four-categorical account

6.2.3 Moments

The word *Moment* is derived from the german *Momente* in the writings of Husserl and it denotes, in general terms, what is sometimes named *trope* (Williams, 1966; Schneider, 2003b), *abstract particular* (Stout, 1921; Campbell, 1990), *mode* (Lowe, 2001; Schneider, 2002), *particular quality*, *individual accident*, or *property instance*. In the scope of this work, the term bears no relation to the notion of time instant in ordinary parlance. The origin of the notion of moment lies in the theory of individual accidents developed by Aristotle in his Metaphysics and Categories. For him, an accident is an individualized property, event or process that is not a part of the essence of a thing. We here use the term "moment" in a more general sense and do not distinguish *a priori* between essential and inessential moments.

As pointed out by (Schneider, 2002), there is solid evidence for moments in the literature. On one hand, in the analysis of the content of perception (Lowe, 2001, p. 205; Mulligan & Simons & Smith, 1984, p. 304-308), moments such as colours, sounds, runs, laughter and singings are the immediate objects of everyday perception. On the other hand, the idea of moments as truthmakers (Mulligan & Simons & Smith, 1984, p. 295-

304) underlies a standard event-based approach to natural language semantics, as initiated by Davidson (1980, pp. 118-119) and Parsons (1990, chaps. 1-3).

The notion of moment employed here comprises:

1. *Intrinsic Moments*: qualities such as a color, a weight, a electric charge, a circular shape; modes such as a thought, a skill, a belief, an intention, a headache, as well as dispositions such as the refrangibility property of light rays, or the disposition of a magnetic material to attract a metallic object;

2. *Relational Moments (or relators)*: a kiss, a handshake, a covalent bond, but also *social objects* such as a flight connection, a purchase order and a commitment or claim (Wagner, 2003; Guizzardi & Wagner, 2005a).

An important feature that characterizes all *moments* is that they can only exist in other individuals (in the way in which, for example, electrical charge can exist only in some conductor). To put it more technically, we say that moments are *existentially dependent* on other individuals (see definition 5.10, in chapter 5), named their *bearers*. Existential dependence can be used to differentiate intrinsic and relational moments: intrinsic moments are dependent of one single individual; relational moments depend on a plurality of individuals.

Existential dependency is a necessary but not a sufficient condition for something to be a moment. For instance, the temperature of a volume of a gas depends on, but is not a moment of its pressure. Thus, for an individual x to be a moment of another individual y (its bearer), a relation of *inherence* – sometimes called *ontic predication* – must hold between the two, symbolized as i(x,y). For example, inherence glues your smile to your face, or the charge in a specific conductor to the conductor itself. In summary, moments are *ways things are* (Armstrong, 1989; Lowe, 2001) and, hence, cannot be conceived independently of the particulars they inhere in.

Inherence is an irreflexive, asymmetric and intransitive relation between moments and other types of endurants. These formal properties are represented in the formulas (5), (6) and (7) below, respectively. Additionally, as expressed in formula (4), inherence is a special type of existential dependence relation between particulars (symbolized as *ed* below):

(3). $\forall x,y\ (i(x,y) \rightarrow \mathbf{Moment}(x) \wedge \mathbf{Endurant}(y))$
(4). $\forall x,y\ (\mathrm{i}(x,y) \rightarrow \mathrm{ed}(x,y))$
(5). $\forall x\ \neg \mathrm{i}(x,x)$
(6). $\forall x,y\ (\mathrm{i}(x,y) \rightarrow \neg \mathrm{i}(y,x))$

(7). $\forall x,y,z\ (i(x,y) \wedge i(y,z) \rightarrow \neg i(x,z))$

According to (3), moments inhere in other endurants, which can themselves be moments. An example of moment inhering in another moment is the radius of a circular form. The infinite regress in the inherence chain is prevented by the fact that there are endurants that cannot inhere in other individuals, namely, *substantials* (see section 6.2.4). We can, thus, formally characterize a moment as an individual that inheres in (and, hence, is existentially dependent upon) another individual:

Definition 6.1 (Moment): A moment is an endurant that *inheres in*, and, therefore, is *existentially dependent* of, another endurant. Formally,

(8). $\mathbf{Moment}(x) =_{def} \mathbf{Endurant}(x) \wedge \exists y\ i(x,y)$

■

In our framework we adopt the so-called *non-migration* (Guizzardi & Herre & Wagner, 2002a) or non-*transferability* (Martin, 1980) *principle*. This means that it is not possible for a moment *m* to inhere in two different individuals *a* and *b*:

(9). $\forall x,y,z\ (\mathbf{Moment}(x) \wedge i(x,y) \wedge i(x,z) \rightarrow y = z)$

If a moment x inheres in an individual y, y is termed the *bearer* of x and is defined as follows:

Definition 6.2 (Bearer of a Moment): The bearer of a moment x is the unique[48] individual y such that x *inheres in* y. Formally,

(10). $\beta(x) =_{def} \iota y\ i(x,y)$

■

This characteristic of moments seems at first counterintuitive. For example, if we have two particulars *a* (a red apple) and *b* (a red car), and two moments m_1 (particular redness of *a*) and m_2 (particular redness of *b),* we consider m_1 and m_2 to be different individuals, although perhaps qualitatively indistinguishable. What does it mean then to say that *a* and *b* have the *same* color? Due to (9), sameness here cannot refer to strict (numerical) identity, but only to a qualitative one (i.e., equivalence in a

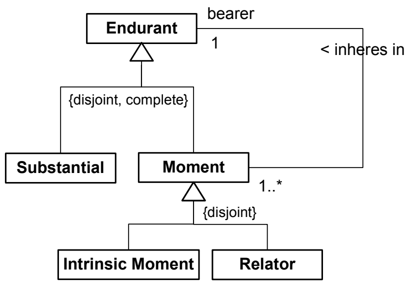
---

[48]The iota operator (ι) used in a formula such as $\iota x \varphi$ was defined by Russel in (Russel, 1905) and implies both the existence and the uniqueness of an individual x satisfying predicate φ.

certain respect). In conformance with DOLCE (Masolo et al., 2003a), we distinguish between the color of a particular apple (its quality) and its 'value' (e.g., a particular shade of red). The latter is named *quale*, and describes the position of an individual quality within a certain *quality dimension*. The notions of quale and quality dimension are discussed in depth in section 6.2.6.

Figure 6.4 depicts the inherence relation between moments and their bearers. Relators and modes are further discussed in section 6.2.7.

Figure 6-4 Moments and their unique bearers



6.2.4 Substantials

In the previous section, we have formally stated that moments inhere in other individuals, forming a chain of inherence that ends in a *substantial*. Since inherence is a sort of existential dependence, we have that all moments are existentially dependent on other individuals. Substantials, in contrast, by not inhering in anything, enjoy a higher degree of independence. We, thus, define the category of substantials as follows.

Definition 6.3 (Substantial): A substantial is an endurant that does not *inhere in* another endurant, i.e., which is not a moment. Formally,

(11). Substantial(x) $=_{def}$ Endurant(x) $\land \neg$Moment(x)

■

Substances are individuals that posses (direct) spatial-temporal moments and are founded on matter. They can be further classified in either *Amounts of matter* or *Objects*. The distinction is made based on whether members of these categories satisfy a unity condition. *Amounts of Matter* (are also known as Stuff) are substantials with no unity; all their parts are essential, i.e., they are mereologically invariant: the identity of an amount of mater is determined by the sum of its parts and, thus, a change in one of the parts changes the identity of that particular (see section 5.5.1). Since countability is strongly related to unity (see section 5.8), amounts of matter are individuals with determinate principles of identity but with indeterminate counting principles. (Lowe, 2001) terms these entities *quasi-objects*. Examples of *Amounts of Matter* are individuals linguistically referred by mass nouns such as “sugar”, “sand”, and “gold”.

Objects, conversely, are substantials with unity, i.e., integral wholes unified by a certain unity criteria (see section 5.3). Contrary to amounts of matter, it is not the case that all parts of an object are essential. Instead of necessarily obeying a mereological principle of identity, the principle of identity of objects is supplied by the kinds (substance sortals) that they instantiate. These principles of identity determine which parts are essential, mandatory or merely contingent. Examples of objects include ordinary mesoscopic entities that are linguistically referred to as count nouns, such as a dog, a house, a hammer, a car, Alan Turing and The Rolling Stones but also *Fiat Objects* such as the North-Sea and its proper-parts, postal districts and a non-smoking area of a restaurant (Smith, 1994).

We also consider as objects those parasitic substantials such as stains, edges, bumps, which are named features in DOLCE, but also what (Pribbenow, 2002) terms a *negative object* (e.g., a hole, the interior of a drawer). As a consequence, the relation between these parasitic entities and their hosts is one of *inseparable parthood* (see section 5.4.2) not one of inherence. For instance, we say that a hole in a piece of cheese is an inseparable part of the cheese, as opposed to one of its moments.

Objects can also have parasitic substantials as essential parts. For instance, the key whole of a locker may be considered an essential part of the locker. In (Guizzardi & Wagner, 2005b), we make a further distinction between two types of objects: (i) agents - to which we can ascribe mentalistic notions (mental moments) such as beliefs, desires and intentions, and non-agentive objects. Here, in contrast, we shall not consider this distinction and, thus, we concentrate only on the common properties of objects.

As discussed in chapter 5, conceptual models require that instantiated objects have determinate identities. For this reason, in order to represent *amounts of matter*, we must refer to them in a special sense, namely, that of a maximally-self-connected amount of matter. These objects, termed in chapter 5 *quantity* are genuine objects, possessing both determinate identity and determinate countability.

Figure 6.5 below depicts the types of substantials considered here and their relation to moments.

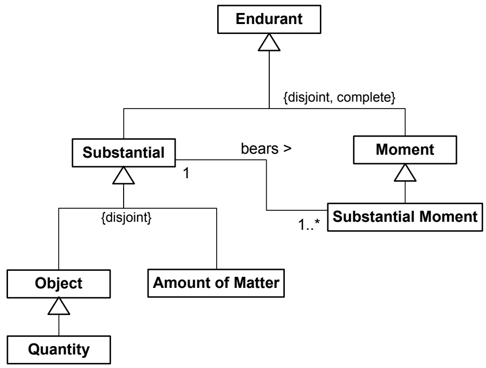


Figure 6-5 Types of Substantials and their relation to moments

Figure 6.5 also illustrates the *bears* relation between a substantial and its moments. This relation is the inverse of the *inheres in* relation shown in figure 6.4. As moments must inhere in some in individual (not necessarily a substantial), substantials must bear some moments, i.e., *"there are no propertyless individuals"* (Bunge, 1977), or bare particulars (see section 4.2). This is expressed in formula (12) below. We name *substantial moments* those qualities that necessarily inhere in a substance.

(12). $\forall \mathbf{x}$ **Substantial(x)** $\rightarrow \exists$**y (Moment(y)** $\wedge$ ***i*****(y,x))**

Finally, in the beginning of this section we state that substantials enjoy a higher degree of independence when compared to moments. Can we make a stronger statement? Can we say that substantials are existentially independent from all other individuals?

If we take the notion of existential dependency that we give in definition 5.10 in chapter 5, the answer is no. Existentially dependence has been defined as follows:

Definition 5.10 (existential dependence): Let the predicate ε denote existence. We have that an individual *x* is *existentially dependent* on another individual *y* iff, as a matter of necessity, y must exist whenever x exists, or formally

$$ed(x,y) =_{def} \Box(\varepsilon(x) \rightarrow \varepsilon(y))$$

■

Now, there are certainly pairs (x,y) where x is a substantial that satisfy ed(x,y). For example, if y is any of the essential moments of x. Moreover, even if both x and y are substantials, ed(x,y) can be satisfied. Take for example a substantial and any of its essential parts. Or, alternatively, a substantial x and another object y of which x is an inseparable part (see definition 5.15). However, we can say that if x and y are two substantials and they are disjoint then they must also be independent from each other (symbolized as *indep*):

(13). $\mathbf{indep(x,y) =_{def} \neg ed(x,y) \land \neg ed(y,x)}$
(14). $\mathbf{\forall x,y\ Substantial(x) \land Substantial(y) \land (x \int y) \rightarrow indep(x,y)}$

For example, a person depends on her brain, and a car depends on its chassis. However, a person (car) does not dependent on any other substance which is disjoint from her (it). Notice that formula (14) also excludes the case of mutual existential dependence between substantials that share a common essential part (see chapter 5).

6.2.5 Universals

To complete the Aristotelian Four-Categories ontology depicted in figure 6.3, we consider the existence of both *substantials universals* and *moment universals*. We use the term universal in a broader sense than it is sometimes used in the philosophical literature, such as for instance in (Armstrong, 1989). Therefore, we do not necessarily commit to existence of universals as *abstract entities that are multiple instantiated in several individuals*. Instead, we employ the term in a sense which is equivalent to term *Category* in (Heller et al., 2004). The position should be made clearer in the course of this section. For now, a universal can be considered simply as something (i) which can be predicated of other entities (or, in the Aristotelian sense, *said of* or attributed to other entities) and (ii) that can potentially be represented in language by *predicative terms*. In summary, as a synonymous of *type* as we have used it in chapter 4.

We use the symbol :: to denote the instantiation relation, a basic formal relation defined to hold between individuals (first argument) and universals (second argument). Hence, when writing x::U we mean that x is an instance of U or that x has the property of being a U:

(15). $\mathbf{\forall x,U\ x::U \rightarrow Individual(x) \land Universal(U)}$

The difference between substantial and moment universals can be characterized as follows (Schneider, 2002):

Definition 6.4 (Substantial and Moment Universals): A Substantial universal is a universal that is instantiated only by substantial individuals. Analogously, a moment universal is a universal that is only instantiated by moment individuals. Formally,

(16). $\mathbf{SubstantialUniversal(U) =_{def} Universal(U) \land}$

$$\forall x\ (x::U \rightarrow \textbf{Substantial}(x))$$

(17). $\textbf{MomentUniversal}(U) =_{def} \textbf{Universal}(U) \wedge \forall x\ (x::U \rightarrow \textbf{Moment}(x))$

■

Although we want to avoid making unnecessary commitments to particular theories of universals, we believe that some interpretations for the nature of universals (and the corresponding theories) must be explicitly ruled out. To begin with, for all the reasons discussed in section 6.2.1, we exclude all sorts of *class and predicate nominalisms* (including the *natural class theory*). These theories are named in (Armstrong, 1989) *Blob theories* because of their failure to account for the existence and priority of properties. In conformance with Armstrong we defend that individuals belong to a certain category because they share a number of properties, not the other way around. For this reason, we also exclude another type of Blob theory that has not been discussed so far, namely, the classical formulation of the *resemblance theory* (Price, 1953).

Traditionally, class theories of universals are extensional theories. Thus, as discussed in section 6.2.1, two classes are identical if they apply to the same individuals. Here, we reject this strong nominalist idea that equates universals with sets. For every universal U there is a set *Ext(U)*, called its *extension*, containing all instances of U as elements. However, even if two universals U_1 and U_2 have identical extensions ($Ext(U_1) = Ext(U_2)$), they are not necessarily considered to be identical.

The identity of universals instead should be analyzed in terms of the identity of the supplied principles of application and identity, and/or in terms of the causal powers bestowed by them (Armstrong, 1989). However, a fuller consideration of this issue falls outside the scope of this thesis. The following definition relates the instantiation relation, universals and their extensions:

Definition 6.5 (Extension of a Universal): The extension of a universal U is the set S that contains all instances of U, and only them. Formally,

(18). $\textbf{Ext}(U) =_{def} \{x \mid x::U\}$

■

In chapter 4, we discuss the ontological distinction between sortals and non-sortals. There is, in particular, an intimate relation between the category of non-sortals named mixins (or attributions) and moment universals. This distinction is also present in Aristotle's original differentiation between what is *said of a subject* (*de subjecto dici*), denoting instantiation and what is *exemplified in a subject* (*in subjecto est*), denoting inherence. Thus, the linguistic difference between the two meanings of the copula "is" reflects an ontological one. For example, the ontological interpretation of the sentence "Jane is a Woman" is that the object Jane *instantiates* the (substantial) universal Woman. However, when saying that "Jane is tall" or "Jane is laughing" we mean that Jane *exemplifies* the moment universal Tall or Laugh, by virtue of her specific height or laugh. Thus, in pace with (Schneider, 2002), besides instantiation, we recognize another formal relation that can obtain between individuals and universals, namely, the relation of *exemplification*:

Definition 6.6 (Exemplification): An individual x exemplifies the moment universal U iff there is a particular moment y instance of U that inheres in x. Formally,

(19). exemplification(x,U) $=_{\text{def}}$ MomentUniversal(U) $\wedge$ $\exists$y (i(y,x) $\wedge$ y::U)

■

In formula (19) above the variable x is restricted to the category of endurants but not to its subcategory of substantials. This is because, since moments can also inhere in moments, the exemplification relation can also exist between moments and moment universals. For example, in "Jane is laughing", Jane's laugh can exemplify the moment universal *frankness* (Schneider, 2002), or in "The ball is red", the color of the ball can exemplify the universal beauty.

As mentioned above, universals are not sets. Differently from sets, they are defined intensionally. According to (Guizzardi & Herre & Wagner, 2002a,b; Guizzardi & Wagner & Herre, 2004), we capture the intension of a universal by means of an axiomatic specification, i.e., a set of axioms that may involve a number of other universals representing its essential features. A particular form of such a specification of a universal U is called an *elementary specification*.

Definition 6.7 (Elementary Specification): An elementary specification of a universal U consists of a number of universals $U_1,\ldots,U_n$ and corresponding functional relations $R_1, \ldots,R_n$ which attach instances from the U_i to instances of U, expressed by the following schema:

(20). $\forall a(a :: U \rightarrow \exists e_1 ... e_n \bigwedge_{i \leq n} (e_i :: U_i \wedge R_i(a, e_i)))$

The universals $U_1,\ldots,U_n$ used in an elementary specification are called *features*. A special case of an elementary specification is a *intrinsic moments specification* where $U_1,\ldots,U_n$ are intrinsic moment universals.

■

The relation between a universal and the features in its elementary specification is one of *characterization*:

Definition 6.8 (Characterization): A universal U is characterized by a moment universal M iff every instance of U exemplifies M. Formally,

(21). $$\mathbf{characterization(U,M) =_{def} Universal(U) \land MomentUniversal(M) \land \forall x\ (x::U \rightarrow \exists y\ y::M \land i(y,x))}$$

■

Finally, there is another sort of *Blob theory* that we have not explicitly excluded yet, namely, the theory of universals known as *Platonic realism*, or simply, *Platonims* (Balaguer, 2004). Platonism amounts, in a nutshell, to the following: (a) universals are self-existent, ideal and external to individuals, and (b) universals precede their instances but can be exemplified by the latter (Bunge, 1977, p.103). Due to the independence of universals from their instances, Platonic realism leaves open the possibility for the existence of universals that have no instances. According to Bunge, *"there is not a shred of empirical evidence for the hypothesis that forms [i.e., universals] are detachable from their carriers [e.g., their bearers]"*. Therefore, we take here an Aristotelian view of universals w.r.t. accepting the so-called *principle of instantiation* (Armstrong, 1989, p.75). This principle states that, in order to exist, universals must have instances:

(22). $$\mathbf{\forall x\ Universal(x) \rightarrow \Diamond \exists y\ y::x}$$

Much of what has been discussed in this section is summarized in figure 6.6 below.

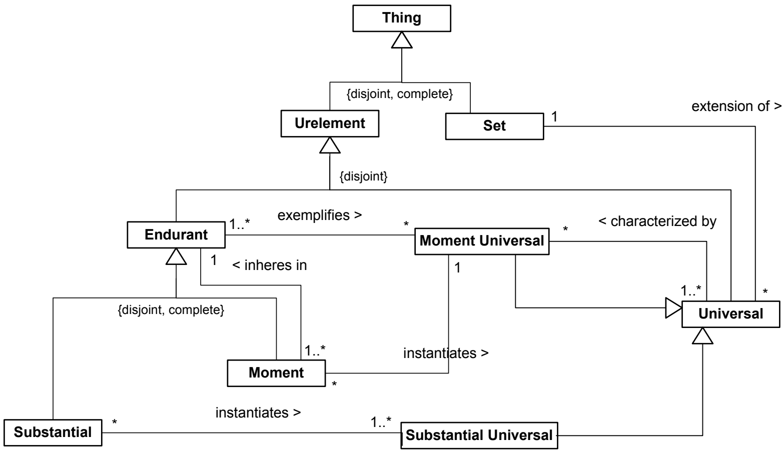


Figure 6-6 Substantials, Moments, Substantial Universals, Moment Universals and their interelations

6.2.6 Qualities, Qualia, Quality Dimensions and Quality Domains

An attempt to model the relation between moments and their representation in human cognitive structures is presented in the theory of *conceptual spaces* developed by the Swedish philosopher and cognitive scientist Peter Gardenfors (Gardenfors, 2000; 2004). The theory is based on the notion of *quality dimension*. The idea is that for several perceivable or conceivable moment universals there is an associated quality dimension in human cognition. For example, height and mass are associated with one-dimensional structures with a zero point isomorphic to the half-line of nonnegative numbers. Other moments such as color and taste are represented by several dimensions. For instance, taste can be represented as a tetrahedron space comprising the dimensions of saline, sweet, bitter and sour, and color can be represented in terms of the dimensions of hue, saturation and brightness. Here, we do not distinguish between physical (e.g., color, height, weight, shape) and nominal moments (e.g., social security number, the economic value of an asset).

According to Gardenfors, some of the quality dimensions (especially those related to perceptual qualities) seem to be innate or developed very early in life. For instance, the sensory moments of color and pitch are strongly connected with the neurophysiology of their perception. Other dimensions are introduced by science or human conventions. For example, the representation of Newton's distinction between mass and weight is not given by the senses but has to be learned by adopting the conceptual space of Newtonian mechanics. Another example is the cognitive dimension of time, which is deemed culture-dependent (while in western cultures we perceive time as a one-dimension linear structure, other cultures perceive time as circular or recurrent). Whereas scientific or conventional dimensions are prescribed by a community of users or defined by a scientific theory, phenomenal quality dimensions have to be empirically extracted via analysis of subject's judgement of similarity between different stimuli (Wilson & Keil, 1999; Shepard, 1962a,b).

As discussed in section 4.3, there is empirical support in the cognitive psychology literature for the thesis that some dimensions are innate and have been hard-wired in the brain for evolutionary reasons. Examples include are our tri-dimensional perception of space and our perception of spatio-temporal continuity. These dimensions are used in our first applied principles of individuation, persistence and identity. On the solid ground provided by these innate dimensions, we expand our conceptual space by including different dimensions associated with the perception of other moments. In fact, Gardenfors explicitly considers learning as the activity of expanding one's conceptual space into new quality dimensions (ibid., p.28) and Quine makes note that some innate quality dimensions are needed in order for learning to be possible (Quine, 1969, p.123).

Gardenfors distinguishes between *integral* and *separable* quality dimensions: "certain quality dimensions are integral in the sense that one cannot assign an object a value on one dimension without giving it a value on the other. For example, an object cannot be given a hue without giving it a brightness value… Dimensions that are not integral are said to be separable, as for example the size and hue dimensions." (Gardenfors, 2000, p.24). He then defines a *quality domain* as "a set of integral dimensions that are separable from all other dimensions (Gardenfors, 2000, p.26)" and a *conceptual space* as "collection of one or more domains" (ibid.). Finally, he defends that the notion of conceptual space should be understood literally, i.e., quality domains are endowed with certain geometrical structures (topological or ordering structures) that constrain the relations between its constituting dimensions. In his framework, the perception or conception of a moment individual can be represented as a point in a quality domain. In accordance with DOLCE (Masolo et al., 2003a) and (Goodman, 1951), this point is named here a *quale*.

We adopt in this work the term *quality structures* to refer to quality dimensions and quality domains. Additionally, we use the term *quality universals* for those moment universals that are associated with a quality structure. Finally, we name *quality* a moment individual that instantiates a quality universal. In the sequel we define these notions formally. We use the predicates QS, QDom and QDim for quality structure, quality dimension and quality domain, respectively.

(23). $\forall x\ \mathbf{QS}(x) \leftrightarrow \mathbf{QDom}(x) \lor \mathbf{QDim}(x)$

Definition 6.9 (Quality, Quality Universal and the association relation): We define the formal relation of association (*assoc*) between a quality structure and an intrinsic moment universal. A quality universal is an intrinsic moment universal that is associated with a quality structure, i.e.,

(24). qualityUniversal(U) $=_{def}$ **intrinsicMomentUniversal(U)** $\wedge$ $\exists!$**x QS(x)** $\wedge$ **assoc(x,U)**

Qualities are the instances of quality universals:

(25). quality(x) $=_{def}$ $\exists!$**U qualityUniversal(U)** $\wedge$ **(x::U)**

Finally, quality structures are always associated with a unique quality universal

(26). QS(x) $=_{def}$ $\exists!$**U qualityUniversal(U)** $\wedge$ **assoc(x,U)**

■

An example of a quality domain is the domain of tone with the dimensions of pitch and loudness. Another example is the set of integral dimensions related to color perception, which is further explored in the sequel.

A color quality *c* of an apple *a* takes it value in a three-dimensional color domain constituted of the dimensions hue, saturation and brightness. Figure 6.7 depicts the geometric space spawned by the three quality dimensions that form this domain. The *hue* dimension stands for a term whose meaning is close to that of the word color in everyday life. It is a characteristic of the co-called *chromatic* colors (e.g., red and blue) but not of the achromatic colors (i.e., black, white and the totality of neutral greys in between). In figure 6.7, this quality dimension is represented as a circle and the polar coordinates describing the angle of the color around the circle gives the hue value.

Saturation (or Chromaticness) refers to the "purity" of the color, that is, the degree to which it is chromatic instead of achromatic. The more grey (or black and white) is mixed with a color, the less saturation it has. In figure 6.7, this quality dimension has a representation that is homomorphic to the real line ranging from grey (zero color intensity) to increasingly greater intensities. Figure 6.6 depicts the color circle view of the dimensions of hue and saturation together.

Brightness is a measure that varies both among the chromatic and achromatic colors, although it stands out most clearly in the latter. In figure

6.7, brightness is represented as linear quality dimensions with two end points, namely white and black. In other words, White is hueless and maximally bright; Black is hueless and minimally bright (Gleitman, 1991).

Some points in figures 6.7 and 6.8 are worth mentioning regarding the geometry of these representations:

– In figure 6.8 one can easily observe the relation of *complementariness* between colors. Orange can be seen as the overlapping region between Red and Yellow, and Violet as overlapping between Red and Blue. The Red region can therefore be accounted as the sum of the Orange, Violet and (what is termed in color theory) *unique Red*. Likewise, the Green region in the color space is the sum of unique Green, Yellowish green and Blueish green (Fischer Nilsson, 1999). Since Red and Green do not overlap they are said to be *complementary colors*. The same holds for Yellow and Blue. Quality dimensions, quality regions, quality domains and conceptual spaces are one of the best examples of entities related by a type of parthood that obeys the full axiomatization of the Classical Extensional Mereology (see chapter 5). In fact, in DOLCE (Masolo et. al., 2003a), quality regions are defined as mereological sums of qualia, and a *quality space* (a notion analogous to conceptual space) as a mereological sum of quality regions. Finally, from a pragmatics point of view (see section 2.2.3), the visual geometric representation of conceptual spaces represent a case of *pragmatic systematicity* and afford cases of the so-called inferential *free-rides* (Shimojima, 1996);

– The notion of opposition (complement) is derived from the geometrical structure generated by the color (hue) circle (e.g., it is meaningless to talk about opposite weights). In the linear dimensions of saturation and brightness, conversely, an ordering relation can be defined. For instance, the ordering {minimum (black) < low < medium < high < maximal (white)} can be defined for regions of the brightness quality dimension (Fischer Nilsson, 1999). Likewise, the axiomatization of *total orders* can be defined for the dimensions of weight, height, age, etc. In section 6.4.3, we show how the structure of a quality dimension Q can be used to derive characteristics for the so-called *comparative formal relations* that are based on Q;

– The geometric structure of figure 6.7 constrains the relation between some of these dimensions. In particular, saturation and brightness are not totally independent, since the possible variation of saturation decreases as brightness approaches the extreme points of black and white, i.e., for almost black or almost white, there can be very little variation in saturation. A similar constraint could be postulated for the relation between saturation and hue. When saturation is very low, all hues become similarly approximate to grey.

Figure 6-7 The Quality Dimensions of Hue, Saturation and Brightness forming the Color Splinter (Gardenfors, 2000)

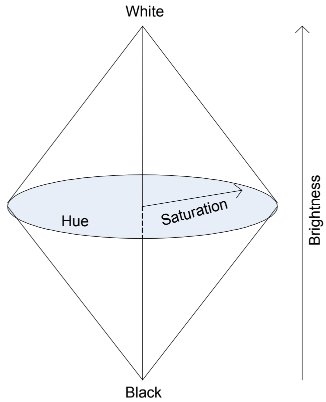
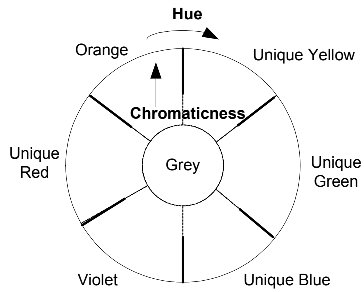


Figure 6-8 The Color Circle view of the dimensions of Hue and Saturation



The trichromatic color quality domain is closely related to the human physiology of color perception. In the animal kingdom, a variety of color systems can be found (Thompson, 1995). For instance, many mammals are dichromats, while others are tetrachromats (e.g., goldfish and turtles). Additionally, for some specific artificial systems, the color domain can be specified in terms of different quality dimensions (e.g., the RGB system) (Raubal, 2004). The important point to be emphasized here is the following: for the same quality universal, there can be potentially many conceptual spaces associated with it, and which one is to be adopted depends on the objectives underlying each a specific conceptualization.

A similar point is made in the GOL ontology (Heller et al., 2004). In GOL, there is an explicit distinction between what is termed a *property* (e.g., size) and what is termed a *property value* (e.g., small, medium, big) of that property. The idea is similar to the one defended here that for a quality universal (GOL-property) there can be a *measurement structure* associated with it. For a given property P of a given individual x there is an *instrument* I that associates P with a "value" in a measurement structure. This value can be a point or a *region* of the measurement structure.

The GOL notion of a measurement structure is akin the notions of quality domain (or quality dimension as a border case of quality domain) as used here. Likewise, the GOL's property value can be interpreted as our notion of quale. In summary, in conformance with GOL, we propose that for a given quality universal there can be different quality domains associated with different *measurement instruments*. In particular, human perceptual systems can be seen as an example, in which case the quality regions roughly correspond to qualitative sensorial experiences of humans. We also consider social conventions as instruments that are able to generate abstract quality dimensions and domains for nominal quality universals such as address, social security number, license place, etc.

Once more, the notion of *quality region* should be taken here literally, i.e., as spatial regions determined by the geometry or the topology of a conceptual space. For instance, in a one-dimensional quality domain, say the time dimension, the time point *now* divides this dimension into two regions the *past* and *future* regions, which are both one-dimensional regions (lines) of the time domain obeying the same ordering axiomatization of the entire dimension. Conversely, in a bidimensional domain represented in a cartesian space, a region is a sub-area of the domain. Most quality domains are *metric spaces* (Gardenfors, 2000). The concepts of quale, metric space, distance and region are formally defined as follows.

Definition 6.10 (quale): A point in a n-dimensional quality domain can be represented as a vector $v = \langle x_1 \ldots x_n \rangle$ where each x_i represents each of the integral dimensions that constitute the domain. A multidimensional quale is therefore the vector representing the several quality dimensions that are mutually dependent in a quality domain.

■

Definition 6.11 (metric space) (Weisstein, 2004)**:** Let S be a quality domain (set of qualia). A metric space is obtained by associating with S a distance function *d* (called the metric of S) such that, for every two quality values x,y in S, d(x,y) represents the distance between x and y in the domain S. The distance function obeys the following constraints:

a) $d(x,y) = 0$ iff $(x = y)$;
b) $d(x,y) = d(y,x)$;
c) $d(x,y) + d(y,z) \geq d(x,z)$ (triangle inequality)

Generally, the similarity between two vectors x = $\langle x_1 \ldots x_n \rangle$ and y = $\langle y_1 \ldots y_n \rangle$ in a n-dimensional domain can be computed by the *Minkowskian Metric* (Wilson & Keil, 1999).

$$d(x,y) = \left[\sum_{i=1}^{n} | x_i - y_i |^r \right]^{(1/r)}$$

■

In cognitive structures such as quality domains, the distance function is interpreted as being inversely related to the notion of conceptual *similarity* between two individuals (Wilson & Keil, ibid.). An example of distance function for quality domains (i.e., among integral dimensions) that suitability finds strong empirical evidence in the literature is the *Euclidean Metric* (i.e., for r = 2). In this case,

$$d(x,y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2} \ .$$

■

The definition of a distance function is very important for many reasons, both in theoretical and in practical terms. Given a quality domain and its constituting dimensions, the quality function represents the degree of similarity between two individuals. One can for example give a precise account for why a certain shade of unique red is more similar to orange than to a shade of green, or why the taste of hazelnut is more similar to the taste of walnut than to the one of limes. The other reason is that, as shown by (Gardenfors, 2000), given a number of *prototype* points in a metric quality domain, and a similarity function, a partition of the domain in different quality regions can be generated. Moreover, all generated regions are *convex regions*. The notion of a convex region in a conceptual space can then be used to define the notion of a *quality region*:

Definition 6.12 (quality region): A quality region is a convex region C of a quality domain. A region C is convex iff: for all two points x,y in C, all points between x and y are also in C (Weisstein, 2004).

■

For example, take the color circle generated by the dimensions of hue and saturation in the color domain. By selecting certain color prototypes (for example, empirically) and by applying a distance metric to calculate the similarity between points in this circle, a so-called *Voronoi tessellation* (Weisstein, 2004) is generated, partitioning the circle in the familiar color regions. Moreover, all these regions are indeed convex[49]. Therefore, properties such as Red and Green, which are determinables for the determinate moment universal color, correspond to convex regions in this domain.

It is well understood in philosophical ontology that one should differentiate between the property that things have and the perceptions we have of these properties according to certain measurement instruments employed. Bunge (1977), for instance, uses the term *substantial property* for the ontological entity and the term *attribute* (or predicate) for its logical or linguistic counterpart, and emphasizes the lack of direct correspondence between the two groups of entities. On one hand, there can be properties of things that do not find representation on current measurement systems. On the other hand, predicates can be freely created in conceptual systems and in language that do not represent any *substantial property*. Examples include negation of predicates or predicates formed by logical disjunction, e.g., while one can use the predicate $\neg$horse(x), there is nothing that has the negative property of *not-being-a-horse*. Likewise, while one can use a predicate equivalent to (car(x) $\vee$ plane(x)), there is nothing that has the optional property of *being-a-plane-or-a-car*.

According to (Gardenfors, ibid.), only attributes representing substantial properties will form quality regions in a conceptual space. Borrowing from David Lewis' terminology (1986), we name these attributes *natural attributions*, as opposed to *abundant attributions*. This distinction, which has been unfortunately obliterated in terminological logics-based languages (e.g., OWL), is of great conceptual importance. Abundant attributions are, from a cognitive point of view, very poor sources of inductive knowledge (see section 5.2.3 on a similar argument about mereological sums).

The notion of quality dimensions presented here does not require the dimensions to be dense sets. An example of a discrete quality dimension is a collection of qualia that form a graph (see figure 6.9). Still in this dimension, it is possible to define the distance function as the shortest path between two elements in the graph[50]. Consequently, we still have a measurement of similarity between elements, and we can still generate convex quality regions. Finally, there can be quality dimensions that are not metric (e.g., enumerations). In fact, typically, for the same quality domain,

---

[49]In the case of color circle the relevant notion of line is not one of a familiar Euclidian straight line. This is because the distance metric used (since it is a circle) is a polar distance function not a Euclidean one. For this reason, the lines in this metric space are actually arcs from an Euclidian perspective.

[50]Also in this case, the notion of "line" is one of a path in the graph. In general, line in a conceptual space is a metaphor whose Euclidean appearance depends on the nature of the metric used.

many different conceptual modeling representations can be generated. For instance, from the color spline of figure 6.7, the following representations could be derived (among others):

1. an unstructured enumeration containing only the lexical representatives of color quality regions;
2. a UML datatype with attributes x,y and z (whereas x is a polar coordinate for hue and y and z belong to an interval of real numbers), together with a set of constraints on the possible values that the triplets ‹x,y,z› can assume;
3. a concept lattice in which each node represents a quality region and the arcs represent the topological relations between these regions (e.g., similarity and complementariness).

In other words, depending on the perspective and also on the need of accuracy for a given representation, for a given quality moment m, its corresponding quale can be represented in conceptual models, for example, as a vector of quale values (e.g., a particular shade of red with specific hue, saturation and brighness components) or a quality region (e.g., the red region in the color spline).

In (Fischer Nilsson, 1999), the author proposes a *conceptual space logic* (an extended concept lattice formalism) as a way to represent conceptual spaces that is conformant with the traditional knowledge representation paradigm and that, hence, can be used for deriving datatype specifications in languages such as OWL and RDF. A hue lattice using Nilsson's formalism is depicted in figure 6.9. By combining this lattice with others representing the saturation and brightness dimensions, other formal definitions can be made. For instance, the *pink* region can be defined as the intersection between the red region in the hue dimension with the *high* region in the brightness dimension (pink = red × saturation(weak + strong) × brightness(high)[51]). *Shocking pink* can be defined as red × saturation(strong) × brightness(high) and the region of *warm colors* as (red + yellow + yellowishgreen) × saturation(weak + strong). From the lattice it can be automatically inferred, for instance, that green and red, and blue and yellow are pairwise complementary (green × red = blue × yellow = ⊥), and that "shocking pink is pink" (shocking pink < pink).

---

[51] The operators of + and × correspond, in an extensional perspective, to the set-theoretic operators ∪ and ∩, respectively. The symbols ⊤ and ⊥ are named the *Top* and *Bottom* elements in Lattice theory, respectively. The binary operation γ(X) named the *Peirce Product* can be understood as the restriction of ⊤ to those elements that are related via γ to an element in X. For details, one should refer to (Fischer Nilsson, 1999).

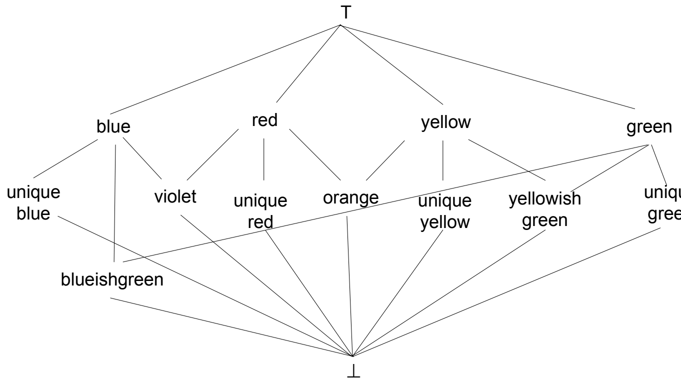


Figure 6-9 A Concept Lattice representing Quality Regions in the Hue dimension and some relations among them

From a metaphysical point of view, quality dimensions and the relations between them, as well as quality domains are *"theoretical entities that can be used to explain and predict various empirical phenomena concerning concept formation"* (Gardenfors, 2000, p.31), i.e., abstract theoretical entities. As a tool for ontological analysis, they provide an interesting *conceptualist*[52] interpretation for W. E. Johnson's notions of *determinable* and *determinate* (Johnson, 1921): every quality region in a quality domain represents a determinable for all its subregions, which in turn, represent its determinates. Since determination is a relative notion (e.g., scarlet is a determinate of red, which is a determinate of color) we should define the ultimate determinable (super-determinable) and the ultimate determinate (super-determinate) in the determination chain (Funkhouser, 2004). In the theory presented here, super-determinables and super-determinates are represented by quality domains and its member qualia, respectively.

From a modeling perspective, the position defended here is that the notion of a quality domain (and the constraints relating different quality dimensions captured in its structure) can provide a sound basis for the conceptual modeling representations of the corresponding quality universal, constraining the possible values that its attributes can assume. This point is further discussed and illustrated in section 6.4.2. In the remainder of this chapter, in order to conform to the representation tradition in conceptual modeling languages, we also take qualia to be abstract entities and represent quality dimensions as *sets of qualia*. Moreover, according to definition 6.11, we take quality domains to be defined in terms of the cross-product of their constituent quality dimensions, and stipulate that the formation rule for the tuples that are members of a quality domain must obey the constraints that relate its quality dimensions. For instance, the *mass* domain can be represented as a subset of the set of Real numbers

[52]In a nutshell, *conceptualism* is the doctrine that equates universals to conceptual categories in cognition. For details, one should refer to (Cocchiarella, 1986; Gardenfors, 2000).

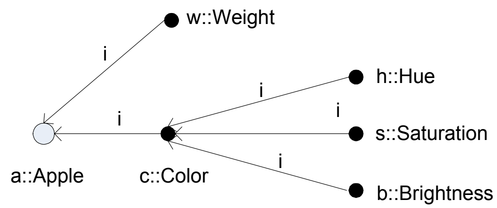
(respecting the same axiomatization) and and the *hue* dimension can be represented as an enumeration of color qualia augmented with a set of formal relations between its members (e.g. *complementaryOf* and *closeTo*). Finally, the color domain can be defined as ColorDomain $\subset$ HueDimension $\times$ SaturationDimension $\times$ BrightnessDimension.

Thus, in this work, quality structures are non-empty sets:

(27). $\forall x\ QS(x) \rightarrow Set(x) \wedge (x \neq \emptyset)$

Following (Masolo et al., 2003a), we take that whenever a quality universal U is related to a quality domain D, then for every individual quality x::U there are *indirect qualities* inhering in x for every quality dimension associated with D. For instance, for every particular quality c instance of Color there are quality individuals h, s, b which are instances of quality universals Hue, Saturation and Brightness, respectively, and that inhere in c. The qualities h, s, b are named *indirect qualities* of c's bearer. This idea is illustrated in figure 6.10 below.

Figure 6-10 Example of an inherence chain from (indirect) quality to quality to substantial



Qualities such as h, s, b and w in figure 6.10 are named *simple qualities*, i.e., qualities which do not bear other qualities. A quality such as c in this figure, in contrast, is named a *complex quality*. The quality universals instantiated by simple and complex qualities are named *simple quality universals* and *complex quality universals*. Formally,

Definition 6.13 (Simple and Complex qualities, and simple and complex quality universals):

(28). $simpleQuality(x) =_{def} quality(x) \wedge \neg\exists y\ i(y,x)$
(29). $complexQuality(x) =_{def} quality(x) \wedge \neg simpleQuality(x)$
(30). $simpleQualityUniversal(U) =_{def} qualityUniversal(U) \wedge (\forall x\ (x::U) \rightarrow simpleQuality(x))$
(31). $complexQualityUniversal(U) =_{def} qualityUniversal(U) \wedge (\forall x\ (x::U) \rightarrow complexQuality(x))$

■

Since the qualities of a complex quality x::X correspond to the quality dimensions of the quality domain associated with X, then we have that no two distinct qualities inhering a complex quality can be of the same type

(32). $\forall x,y,z,Y,Z\ \mathbf{complexQuality}(x) \land (y::Y) \land i(y,x) \land (z::Z) \land i(z,x) \land (Y = Z) \rightarrow (y = z)$

For the same reason, since there are not multidimensional quality dimensions, we have that complex qualities can only bear simple qualities:

(33). $\forall x\ \mathbf{complexQuality}(x) \rightarrow (\forall y\ i(y,x) \rightarrow \mathbf{simpleQuality}(y))$

Moreover, we have that simple quality universals are always associated with quality dimensions and vice-versa, i.e.,
(34). $\forall x,y\ \mathbf{assoc}(x,y) \rightarrow (\mathbf{QDim}(x) \leftrightarrow \mathbf{simpleQualityUniversal}(y))$

and that complex quality universals are always associated with quality domains

(35). $\forall x,y\ \mathbf{assoc}(x,y) \rightarrow$
$(\mathbf{QDom}(x) \leftrightarrow \mathbf{ComplexQualityUniversal}(y))$

Suppose that D is quality domain associated with a (complex) quality universal U, then D can be defined in terms of the cross-product of the quality dimensions associated with the quality universals characterizing U. Formally,

(36). $\forall U,x\ \mathbf{QDom}(x) \land \mathbf{assoc}(x,U) \rightarrow \exists y_1...y_n \exists z_1...z_n\ (x \subseteq y_1 \times ... \times y_n)$
$\bigwedge_{i \leq n} (\mathbf{assoc}(y_i,z_i) \land \mathbf{characterization}(U,z_i)) \land$
$(\forall w\ \mathbf{characterization}(U,w) \rightarrow \bigvee_{i \leq n} (w = z_i))$

We use predicate *ql(x,y)* to represent the formal relation between a quality individual y and its quale x:

(37). $\forall x,y\ \mathbf{ql}(x,y) \rightarrow \mathbf{quale}(x) \land \mathbf{quality}(y)$
(38). $\forall x,y\ \mathbf{ql}(x,y) \rightarrow \exists U\ (y::U) \land \exists D\ \mathbf{assoc}(D,U) \land (x \in D)$

Relation *ql* is total functional for qualities

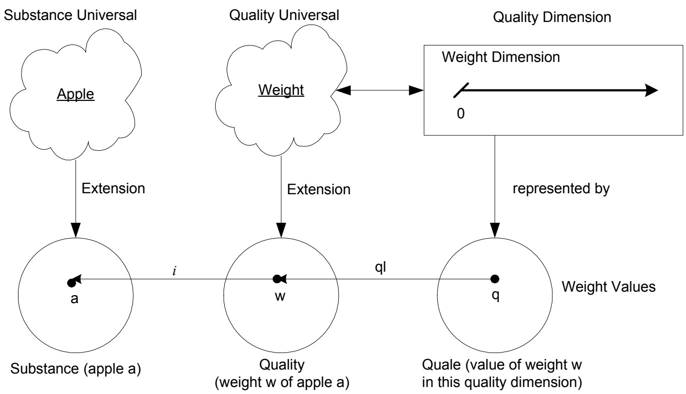
(39). $\forall x\ \mathbf{quality}(x) \rightarrow \exists! y\ \mathbf{ql}(y,x)$

Finally, we require every quale to be a member of a unique quality structure:

(40). quale(x) ↔ ∃!y QS(y) ∧ (x ∈ y)

We summarize the discussion of this section in the following way: suppose we have two distinct particular substantials *a* (a red apple) and *b* (a red car), and two qualities q_1 (particular color of *a*) and q_2 (particular color of *b*). When saying that *a* and *b* have the same color, we mean that their individual color qualities q_1 and q_2 are (numerically) different, however, they can both be mapped to the same point in the color quality domain, i.e., they have the same quale. The relations between a substantial, one of its qualities and the associated quale are summarized in figure 6.11.

Figure 6-11
Substantials, Qualities and Qualia



Quality universals can, in principle, form subsumption taxonomies as much as their substantials counterparts. This is very well reflected in Johnson's notions of *derterminable* and *determinate*. As we mentioned before, determinates can be understood as restrictions of determinables, providing a higher degree of specificity. For instance, *being colored* is a determinable and *being red* is a determinate from it. However, *being red* can also be considered a determinable for *being scarlet* (one of its determinates). In pace with (Heller et al., 2004), we choose to specify *quality universals* here only at the super-determinable level. Moreover, as with *GOL-Properties*, the *determinates* of our moment universals depend on a particular measurement systems and its associated conceptual space. For instance, an individual color quality is an instance of the (super-determinable) universal *Color*. In a given quality domain HSBColorDomain, the relation *ql(c,q)* holds between a quality q and a determinate c (a specific shade of a color in

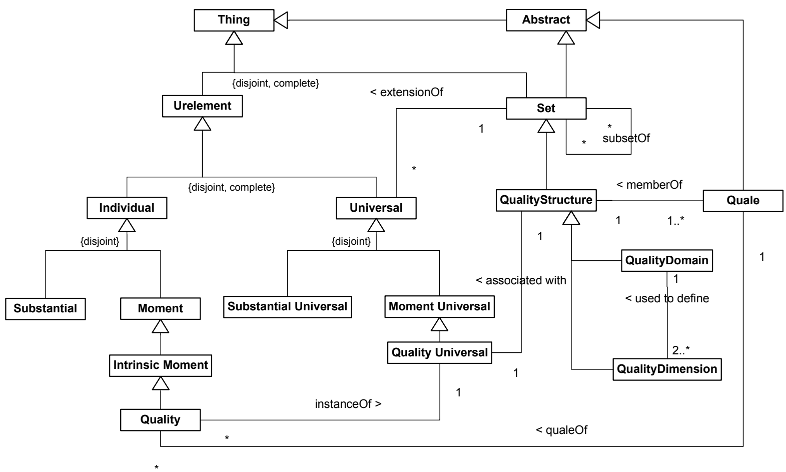
HSBColorDomain), meaning that the quale c is the color value of q in that quality domain.

This is far from disregarding the existence of genuine taxonomies of quality universals. We take here a *realist* stance on universals, in the sense that we believe that universals exist in reality independently of our capacity to know about them. However, we also believe that humans, as cognitive subjects, grasp universals by means of concepts that are in their minds and sometimes cannot capture the universals completely, but only as approximate views. Thus, we defend that, for the purpose of conceptual modeling of things, we must take explicitly into account the conceptual spaces that we use to build the concepts of universals employed in our representations.

Approximating the determinates that we have in our conceptual spaces to the real taxonomy of determinates related to a quality kind is the very task of science discovery, learning and conceptual development (Gardenfors, 2000). In summary, although we believe that every quality instantiates many quality universals related in a determinable/determinate taxonomy, what we mean by a *quality universal* here is only the super-determinate universal in that taxonomy. Therefore, here we consider that for each quality individual there is one and exactly one quality universal that it instantiates.

Figure 6.12 summarizes some important points that we have discussed in this section.

Figure 6-12 Qualities, Quality Universals and their associated Quality Dimensions, and Quality Dimensions as (non-empty) sets of Qualia



6.2.7 Relations and Relators

Relations are entities that glue together other entities. Every relation has a number of relata as arguments, which are connected or related by it. The number of a relation's arguments is called its arity. As much as an unary property such as being Red, properties of higher arities such as *being married-to*, *being heavier-than* are universals, since they can be predicated of a multitude of individuals. Relations can be classified according to the types of their relata.

There are relations between sets, between individuals, and between universals, but there are also cross-categorical relations, for example, between urelements and sets or between sets and universals.

We divide relations into two broad categories, called material and formal relations. *Formal relations* hold between two or more entities directly without any further intervening individual. Examples of formal relations are: 5 is *greater than* 3, this day is *part of* this month, and N is *subset of* Q, but also the relations of *instantiation* (::), *inherence* (i), *quale of a quality* (ql), *association* (assoc), *existential dependence* (ed), among others.

In principle, the category of formal relations includes those relations that form the mathematical superstructure of our framework. We name these relations here *basic formal relations* (Heller et al., 2004) or *internal relations* (Moore, 1919-1920). In this case, in conformance with (Armstrong, 1997; Schneider, 2002) we deem the tie (or nexus) between the relata as non-analyzable. However, we also classify as formal those domain relations that exhibit similar characteristics, i.e., those relations of *comparison* such as is taller than, is older than, knows more greek than. We name these relations *comparative formal relations*. As pointed out in (Mulligan & Smith, 1984), the entities that are immediate relata of such relations are not substantials but qualities.

For instance, the relation heavier-than between two atoms is a formal relation that holds directly as soon as the relata (atoms) are given. The truth-value of a predicate representing this relation depends solely on the atomic number (a quality) of each atom and the material content of heavier-than is as it were distributed between the two relata. Moreover, to quote Mulligan and Smith, "once the distribution has been effected, the two relata are seen to fall apart, in such a way that they no longer have anything specifically to do with each other but can serve equally as terms in a potentially infinite number of comparisons".

Material relations, conversely, have material structure on their own and include examples such as employments, kisses, enrollments, flight connections and commitments. The relata of a material relation are mediated by individuals that are called *relators*. Relators are individuals with the power of connecting entities; a flight connection, for example, founds a relator that connects airports, an enrollment is a relator that connects a student with an educational institution. The notion of relator (relational moment) is supported by several works in the philosophical literature (Heller et Herre, 2004; Loebe, 2003; Degen et al., 2001; Mulligan & Smith, 1986; Mulligan & Simons & Smith, 1984; Bacon, 1995; Simons, 1995; Smith & Mulligan, 1983) and, the position advocated here is that they play an important role in answering questions of the sort: what does it mean to say that John is married to Mary? Why is it true to say that Bill works for Company X but not for Company Y? To quote (once more) Mulligan & Smith's article "The relata of real material relations such as hittings and kissings, in contrast, cannot be made to fall apart in this way: Erna's hitting, r, is a hitting of Hans; it is not a hitting of anyone and everyone who happens to play a role as patient of a hitting qualitatively identical with r. Hence the relational core of such relations cannot be shown to be merely formal".

The distinction between formal and material relations in the quotes from Mulligan and Smith makes it analogous to another distinction among relations, namely the one between bonding and non-bonding relations as proposed by (Bunge, 1977)[53]. For Bunge, bonding relations are the ones that alter the history of the involved relata. For example, the individual histories of John and Mary are different because of the relation "John Kisses Mary". The same is not true for the relation "John is taller than Mary". One can have a world (as history) in which Mary does not exist or in which Mary is taller than John and both individual's history are exactly the same.

Perhaps a stronger and more general way to characterize the difference between formal and material relations is based on their foundation. However, before we proceed, there are some important notions that must be defined. We start with the notion of a *mode*.

Modes are intrinsic moments that are not directly related to quality structures. Gardenfors makes the following distinction between what he calls *concepts* and *properties* (Gardenfors, 2004, p.23): "Properties…form as special case of concepts. I define this distinction by saying that a *property* is based on *single domain*, while a *concept* may be based on *several domains*". We claim that only the moment universals that are conceptualized w.r.t. a single domain, i.e., quality universals, correspond to properties in Gardenfors sense. However, there are moments that as much as substantial universals can be conceptualized in terms of multiple separable quality dimensions. Examples include beliefs, desires, intentions, perceptions, symptoms, skills,

[53]The analogy between formal and material relations and bonding and non-bonding ones is only warranted for the case of relations between individuals. Abstract entities have no spatiotemporal qualities and, consequently, no history. However, here we only consider domain relations between particulars.

among many others. We term these entities *modes*. Like substantials, modes can bear other moments, and each of these moments can be qualities referring to separable quality dimensions. However, since they are moments, differently from substantials, modes inhere necessarily in some bearer. We, thus, define modes as follows:

Definition 6.14 (mode): A mode is an intrinsic moment individual which is not a quality.

(41). $\mathbf{mode(x) =_{def} intrinsicMoment(x) \land \neg quality(x)}$

■

A special type of mode that is of interest here is the so-called *externally dependent modes*. Externally dependent modes are individual modes that inhere in a single individual but that are existentially dependent on (possibly a multitude of) other individuals that are independent of their bearers. Take, for instance, *being the father of*. This is an example of a universal property, since it is clearly multiple instantiated. Suppose that John is the father of Paul. According to our view of universals, in this case, there is a particular instance x of *being the father of*, which bears relations of existential dependence to both John and Paul. However, x is not equally dependent on the two individuals, since x is a moment it must inhere in some individual, in this case, John. Of course, we can also imagine that under these circumstances, there is another extrinsic moment, instance of *being the son of* which conversely inheres in Paul but is also existentially dependent on John. Formally we have,

Definition 6.15 (Externally Dependent Mode): A mode x is externally dependent iff it is existentially dependent of an individual which is independent of its bearer. Formally,

(42). $\mathbf{ExtDepMode(x) =_{def} Mode(x) \land \exists y\ indep(y,\beta(x)) \land ed(x,y)}$

■

The *indep* relation defined in (13) is symmetric. The intention of (42) is to capture circumstances in which an object has a moment in virtue of its association to an external entity. By using the relation *indep* in (42) we can exclude several problematic cases, such as: (i) the moment x being dependent on its bearer; (ii) x being dependent on other moments of its bearer (e.g., color of an object can be considered dependent on the object's spatial extension); (iii) x being dependent on the essential parts or essential wholes of its bearer.

In the case of a material externally dependent moment *x* there is an individual *external* to its bearer (i.e., which is not one of its parts or intrinsic moments), which is the foundation of *x*. The notion of foundation can be seen as a type of *historical dependence* (Ferrario & Oltramari, 2004), in the way that, for instance, *being connected to* is founded in an individual *connection*, or *being kissed* is founded on individual *kiss, being punched by* is founded in an individual *punch*, *working at* is founded in a working contract. Since founding individuals are typically perdurants, we refrain from elaborating on the notion of foundation here.

Suppose that John is married to Mary. There are many externally dependent modes of John that depend on the existence of Mary, and that have the same foundation (e.g., a wedding event or a social contract between the parts). These are, for example, all responsibilities that John acquires by virtue of this foundation. Now, we can define an individual that bears all externally dependent modes of John that share the same dependencies and the same foundation. We term this particular a *qua individual* (Masolo et al., 2004, 2005; Odell & Bock, 1998). Qua individuals are, thus, treated here as a special type of *complex externally dependent modes*. Intuitively, a qua individual is "the way an object participates in a certain relation" (Loebe, 2003), and the name comes from considering an individual only w.r.t. certain aspects (e.g., John qua student; Mary qua musician) (Masolo et al., 2004).

The notion of qua individuals is ancient and comes at least from Aristotle[54]. For example in *On Interpretation* he says that someone might be good *qua* cobbler without being good. This problem, that was called by medieval philosophers the problem of *reduplicatio* or the problem of *qualification* (Poli, 1998), was faced also by Leibniz when he formulated the identity principle known as "Leibniz law" (see chapter 4), and by Brentano (Angelelli, 1967). More recently, in contemporary philosophy qua individuals have been employed in order to solve some problems related to theories of *constitution* (Fine, 1982) and theories of *action* (see Anscombe, 1979 and Fine, ibid.). Furthermore, they have been used in the literature to ascribe incompatible properties to the same individual x that result from x's participation in different relations. This is illustrated by the classical example *"Nixon qua quaker is a pacifist, while Nixon qua republican is not"* (Masolo et al., 2005). In an ontology that countenances the notion of qua individuals, this situation can be modelled by having two different modes *Nixon-qua-quaker* and *Nixon-qua-republican* inhering in the substantial Nixon. In this case, we take that whilst the former exemplifies *pacifism*, the latter does not.

[54] See for example (Szabó, 2003) and, for a more historical and deep account (Baek, 1982).

Our notion of *qua individual* is akin to what is termed a *Material Role* in (Heller et al., 2004) and (Loebe, 2003). Moreover, this notion resembles strongly the one of *roles* in (Wieringa & de Jonge & Spruit, 1995), with the difference that Wieringa and collegues do not discuss this type of external dependence of a role. We resume this discussion on roles in section 7.3, in which we elaborate on the relation between the concepts of role proposed by these authors and the one that we have proposed in chapter 4 as well as their interrelationships.

Finally, we can define an aggregate[55] of all qua individuals that share the same foundation, and name this individual a *relator*. Now, let x, y and z be three distinct individuals such that: (a) x is a relator; (b) y is a qua individual and y is part of x; (c) y inheres in z. In this case, we say that x *mediates* z, symbolized by *m*. Formally, we have that:

(43). $\forall x,y\ \mathbf{m(x,y)} \rightarrow \mathbf{relator(x)} \wedge \mathbf{Endurant(y)}$
(44). $\forall x\ \mathbf{Relator(x)} \rightarrow$
$\forall y\ (\mathbf{m(x,y)} \leftrightarrow (\exists z\ \mathbf{quaIndividual(z)} \wedge (z < x) \wedge i(z,y)))$

Additionally, we require that a relator mediates at least two distinct individuals, i.e.,

(45). $\forall x\ \mathbf{Relator(x)} \rightarrow \exists y,z\ (y \neq z \wedge \mathbf{m(x,y)} \wedge \mathbf{m(x,z)})$

In conformance with (Degen et al., 2001), a relator is considered here as a special type of *moment*. Thus, formally, according to definition 6.1, a relator must inhere in a unique individual, i.e., it must have a bearer (see formula 10). We therefore estipulate that the bearer of a relator r is the mereological sum[56] of the individuals that r mediates, i.e.,

(46). $\forall x\ \mathbf{Relator(x)} \rightarrow (\beta(x) = \sigma y\mathbf{m(x,y)})$

Finally, a relation of *mediation* analogous to that of characterization can be defined to hold between relator universals and those universals of their mediated entities:

---

[55] Relators are genuine integral wholes according to the criteria posed in section 5.4, since the relation of common foundation can be used as a suitable characterizing relation. A similar argument can be made in favour of the bearers of relators (see formula 46) in terms of the relation of common mediation.

[56] The notion of mereological sum is defined in chapter 5. In a nutshell, $\sigma x\varphi$ represents the individual x composed by all individuals that satisfy the predicate φ. In the case of formula (46), the bearer of a relator x is the individual y which is composed by all entities mediated by x.

Definition 6.16 (Mediation): The mediation relation holds between a universal U and a relator universal U_R iff every instance of U is mediated by an instance of U_R. Formally,

(47). mediation(U, U_R) $=_{def}$ Universal(U) ∧ RelatorUniversal(U_R) ∧ ∀x (x::U → ∃r r::U_R ∧ m(r,x))

■

A relator universal is a universal whose instances are relators. Relator universals constitute the basis for defining material relations R whose instances are n-tuples of entities. In general, a material relation R can be defined by the following schema.

Definition 6.17 (Material and Formal Relations): Let $\phi(a_1,\ldots,a_n)$ denote a condition on the individuals $a_1,\ldots,a_n$

(48). $$[a_1\ldots a_n]::R(U_1\ldots U_n) \leftrightarrow \bigwedge_{i \leq n} a_i::U_i \wedge \phi\,(a_1\ldots a_n)$$

A relation is called *material* if there is a relator universal U_R such that the condition ϕ is obtained from U_R as follows:

$$\phi(a_1\ldots a_n) \leftrightarrow \exists k\,(k::U_R \wedge_{i \leq n} m(k,a_i))).$$

In this case, we say that the relation R is derived from the relator universal U_R, or symbolically, *derivation(R,U_R)*. Otherwise, if such a relator universal U_R does not exists, R is termed a formal relation.

■

An example of a ternary material relation is *purchFrom* corresponding to a relator universal *Purchase* whose instances are individual purchases. These individual purchases connect three individuals: a *person*, say *John*, an individual *good*, e.g. the book *Speech Acts by Searle*, and a *shop*, say *Amazon*. Thus,

[John, SpeechActsBySearle, Amazon]::$R_{purchFrom}$(Person, Good, Shop).

Since *John::Person, SpeechActsBySearle::Good*, *Amazon::Shop*, and there is a specific purchase relator *p::Purchase* such that

m(p,John) ∧ m(p, SpeechActsBySearle) ∧ m(p,Amazon)

We obtain the following definition for the triple $[a_1, a_2, a_3]$ being a link of the relation universal *purchFrom* between *Person*, *Good* and *Shop*:

$$[a_1,a_2,a_3]::R_{purchFrom}(Person, Good, Shop) \leftrightarrow John::Person \land SpeechActsBySearle::Good \land Amazon::Shop \land \exists p\ (p::Purchase \land m(p,John) \land m(p, SpeechActsBySearle) \land m(p,Amazon)).$$

We can summarize this section as follows:

– we make a fundamental distinction between formal and material relations. Whilst the former hold directly between two entities without any further intervening individual, the latter are induced by mediating entities called relators. Moreover, material relations are founded by material entities in reality, typically perdurants, which are external to their relata. Comparative formal relations, in contrast, are founded in qualities which are intrinsic to the their relata and, hence, can be reduced to relations between these qualities;

– Relators are special types of (relational) moments, i.e., particularized relational properties. Relators are composed of certain externally dependent modes named qua individuals;

– Qua individuals are (potentially complex) externally dependent modes exemplifying all the properties that an individual has in the scope of a certain material relation;

Formal relations shall be treated here as classes of tuples, but which are defined intensionally, as opposed to extensionally. These entities and their interrelationships are exemplified in figure 6.13.

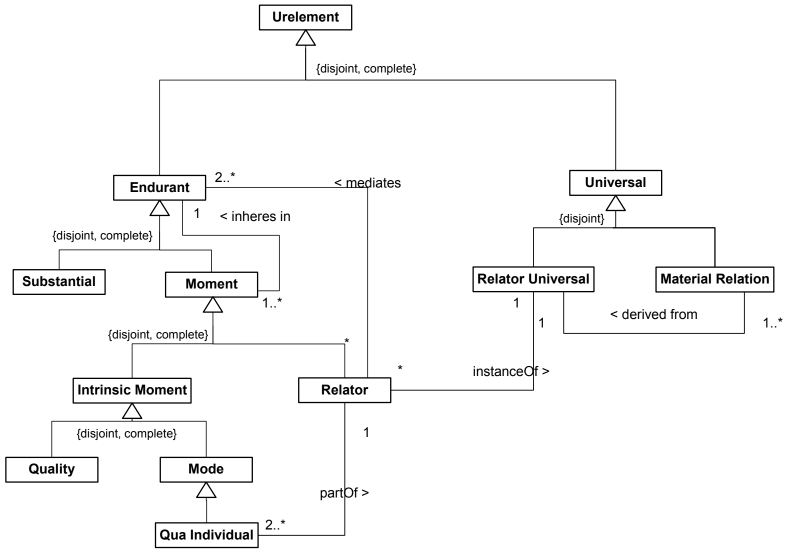


Figure 6-13 Modes, Qua Individuals and Material Relations

6.3 An Ontological Foundation for Conceptual Modeling most Basic Concepts

In this section, we employ the set of ontological categories proposed is section 6.2 to analyze and provide a foundation for some of the most basic constructs in conceptual modeling, namely, *classes*, *objects*, *attributes*, *attribute values* and *associations*. These modeling concepts are represented in practically all conceptual modeling languages. Thus, the conclusions drawn in what follows can be extended to all these languages. However, with the sole purpose of exemplification, we shall refer in the sequel to these concepts as they are represented by UML's modeling primitives. In the remaining of this section, we refer to the *OMG UML Superstructure Specification 2.0* (OMG, 2003c) when quoting text in italics. For simplicity, we write *CM-ontology* when we mean domain ontology in the form of a *conceptual model specification*. Whenever the context is clear, we omit the prefix 'UML' and simply say 'object', 'class', etc., instead of 'UML object', 'UML class', etc.

6.3.1 Classes and Objects

In a UML-based conceptual model specification, an *object* represents a particular instance of a class. It has identity and attribute values (OMG, 2001). A *"Class describes a set of Objects that share the same specification of features,* *constraints and semantics."* (p.86). *"The purpose of a class is to specify a classification of objects and to specify the features that characterize the structure and behaviour of those objects"*. (p.87). Moreover, a UML specification is concerned with describing the *intention* of a class, that is, the rules that define it. The run-time execution provides its *extension*, that is, its instances (OMG, 2001).

We may observe a direct correspondence between universals and classes of a certain kind, as stated in the following principle:

Principle 6.1: In a *CM-ontology*, any universal U of the domain that carries a determinate principle of identity for its instances may be represented as a concrete class C_U. Conversely, for all concrete classes (of a *CM-ontology*) whose instances are basic objects or links (representing individuals), there must be a corresponding universal in the domain.

There are some important points that are represented in this principle. Firstly, it states that only universals that carry a determinate principle of identity for their instances can be represented in a conceptual model as a concrete class (a class that can have direct instances). Therefore, as argued in chapter 4, non-sortal substantial universals must be represented as *abstract classes*. It is a general requirement in conceptual modeling languages that the represented instances must have a definite identity (Borgida, 1990). For this reason, in the category of substantials, what is meant by *object* in conceptual modeling coincides with our use of the term in figure 6.5, i.e., non-object substantials (amounts of matter) can only be represented in a conceptual model as *quantities* (see section 5.5.1).

Moreover, this principle represents an important divergence between the view proposed here and the BWW approach (see, for example, Wand & Storey & Weber, 1999). The proponents of the BWW approach claim that classes in a conceptual model of the domain should only be used to represent substantial universals. In particular, they deny that moment universals should be represented as classes. In our opinion, this claim is not only counterintuitive but also controversial from a metaphysical point of view. This is discussed in depth in section 6.5.1.

Most classes in a *CM-ontology* indeed represent *Substantial Universals*. Firstly, because conceptual models are typically used to model static aspects of a domain and, consequently, the universals represented are typically endurant universals. In addition, Substantials are prior to Moments not only from an existential but also from an identification point of view. For example, (Schneider, 2003b) claims that moments (tropes) are *identificationally dependent* on substantials (objects), i.e., while the latter can be "single out on their own", in order to identify a moment *m* of substantial *s*, one has to identify *s* first.

Principle 6.1, although important to establish the correspondence between conceptual modeling classes and universals, is not elaborated enough to account for the necessary distinctions in this category. For this reason, we defend that the class construct in conceptual modeling should be extended to account for all distinctions among the categories of *substantial* and *moment universals* proposed in chapter 4 of this thesis as well as in the remainder of this section.

6.3.2 Attributes and Data Types

Suppose that we have the situation illustrated in figure 6.11, i.e., a substantial universal *Apple* whose elementary specification contains the feature *Weight*. Thus, for an instance *a* of Apple there is an instance *w* of the quality universal Weight inhering in *a.* The intention of this universal could be represented by the following quality specification:

$$\forall \mathbf{a}\ (\mathbf{a::Apple} \rightarrow \exists \mathbf{w}\ (\mathbf{w::Weight} \land \mathbf{i(w,a)}))$$

Associated with the quality universal Weight we have a quality dimension **WeightValue** and, hence, for every instance *w* of Weight there is a quale *c* denoting a particular weight value, i.e., a point in the weight quality dimension such that *ql(c,w)* holds.

We take the weight quality domain to be a one-dimensional structure isomorphic to the half-line of non-negative numbers, which can be represented by the set **WeightValue**. We assume that any quality domain could be represented as an Algebra A = (V,F), where V is the set of qualia and F a finite set of functions. However, we will not discuss the specifics of algebraic specifications and even less shall we discuss particular algebras. Firstly, because algebraic specification is a topic that has been extensively explored in the literature (see for example Goguen & Malcolm, 2000). Secondly, because here we are only interested in the ontological and cognitive foundations underlying these algebraic specifications. From a conceptual modeling perspective, knowing what should be represented is prior to particular mathematical forms of representation.

The mapping between a substantial *a* and its weight quale can be represented by the following function

$$\mathbf{weight: Ext(Apple)} \rightarrow \mathbf{WeightValue}$$

such that

$$\mathbf{weight(x) = y} \mid \exists \mathbf{z}\ \mathbf{z::Weight} \land \mathbf{i(z,x)} \land \mathbf{ql(y,z)}.$$

In general, let U be a substantial or moment universal and let $Q_1,\ldots,Q_n$ be a number of quality universals. Let E be an elementary specification capturing the intention of universal U:

$$\textbf{(49).}\ \forall x\ (x::U \rightarrow \exists q_1,\ldots,q_n \textstyle\bigwedge_{i\leq n} (q_i::Q_i \wedge i(q_i,x))).$$

If D_i is a quality domain *associated with* Q_i, we can define the function Q_i**:** **Ext(U) → D**$_i$ (named an *attribute function* for quality universal Q_i) such that for every x::U we have that

$$\textbf{(50).}\ Q_i(x) = y \mid y \in D_i \wedge \exists q::Q_i \wedge i(q,x) \wedge ql(y,q).$$

Let us suppose for now a situation in which every Q_i composing in the elementary specification of a universal U (50) is a simple quality universal i.e., Q_i is associated to a quality dimension. In this simplest case, the quality universals appearing in the elementary specification of *U* can be represented in a *CM-ontology* via their corresponding *attribute functions* and associated *quality dimensions* in the following manner:

Principle 6.2: Every attribute function derived from the elementary specification of the universal U may be represented as an attribute of the class C_U (representation of the universal U) in a *CM-ontology*; every *quality dimension* which is the co-domain of one these functions may be represented as data types of the corresponding attributes in this *CM-ontology*. Finally, relations constraining and informing the geometry of a quality dimension may be represented as constraints in the corresponding data type.

For example, in UML *"a data type is a special kind of classifier, similar to a class, whose instances are values (not objects)... A value does not have an identity, so two occurrences of the same value cannot be differentiated"* (p.95). A direct representation of Apple's elementary specification in UML according to principle 6.2 maps the attribute function **weight: Ext(Apple)→WeightValue** to an attribute weight with data type Weight(Kg)[57] in class Apple (figures 6.14.a-b)**.** In UML, specifically,

[57]For now, we shall assume a standard metric unit for each quality dimension (e.g., kilograms in this case). Different metric units affect the granularity but not the structure of a quality domain. For instance, if we consider Weight(Ton) instead of kilograms, the weight dimension is still isomorphic to the half-line of the positive numbers obeying the same ordering.

navigable end names (figure 6.14.b) are a natural alternative mechanism for representing attribute functions since they are semantically equivalent to attributes (OMG, ibid., p.82).

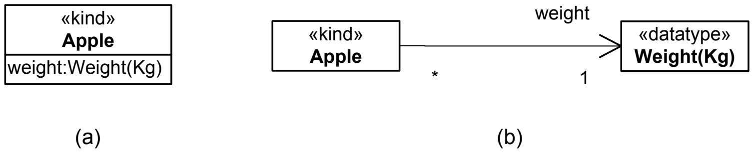


Figure 6-14 Alternative Representations of Attribute Functions in UML

Suppose now that we have the following extension of the elementary specification of the universal Apple represented in figure 6.11:

$$\forall \mathbf{a}\ (\mathbf{a::Apple} \rightarrow \exists \mathbf{c} \exists \mathbf{w}\ (\mathbf{c::Color} \land \mathbf{i(c,a)}) \land (\mathbf{w::Weight} \land \mathbf{i(w,a)}))$$

In order to model the relation between the quality c (color) and its quale, there are other issues to be considered. As previously mentioned, Color is a complex quality universal, and the quality domain associated with it is a three-dimension splinter (see figures 6.7 and 6.8) composed of quality dimensions hue, saturation and brightness. These dimensions can be considered to be *indirect qualities* universals exemplified in an apple *a*, i.e., there are quality individuals h, s, b which are instances of quality universals Hue, Saturation and Brightness, respectively, that inhere in the color quality c (which in turn inheres in substantial a). The intention of the quality universal Color could then be represented by the following specification:

$$\forall \mathbf{c}\ (\mathbf{c::Color} \rightarrow \exists \mathbf{h} \exists \mathbf{s} \exists \mathbf{b}\ (\mathbf{h::Hue} \land \mathbf{i(h,c)}) \land$$
$$(\mathbf{s::Saturation} \land \mathbf{i(s,c)}) \land (\mathbf{b::Brightness} \land \mathbf{i(b,c)}))$$

In this case, we can derive the following attribute functions from the features in this specification:

hue: Ext(Color) → HueValue;
saturation: Ext(Color) → SaturationValue;
brightness: Ext(Color) → BrightnessValue.
Together these functions map each quality of a color *c* to its corresponding quality dimension. One possibility for modeling this situation is a direct application of principle 6.2 to the Color universal quality specification. In this alternative, depicted in figure 6.15, the class Color directly represents the quality universal color and, its attributes the attribute functions *hue*, *saturation* and *brightness*.

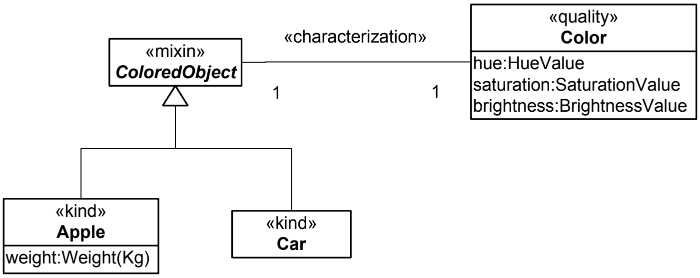


Figure 6-15
Representing Quality Universals and Indirect Qualities

Another modeling alternative is to use directly the construct of a data type to represent a quality domain and its constituent quality dimensions (figure 6.16). That is, we can define the quality domain associated with the universal Color as the set **ColorDomain** $\subset$ **HueValue** $\times$ **SaturationValue** $\times$ **BrightnessValue,** thus complying with (36). Then, we can define the following *attribute function* for the substantial universal Apple according to (50):

color: Ext(Apple) → ColorDomain

such that

$$\mathbf{color(x) = \{\langle h,s,b\rangle \in ColorDomain \mid \exists c{::}Color\ i(c,x) \land (h = hue(c)) \land (s = saturation(c)) \land (b = brightness(c))\}}$$

where hue, saturantion and brightness are the attribute functions previously defined.

Figure 6-16
Representing Qualia in Multi-Dimensional Quality Domain

In figure 6.16, we use the UML construct of a *structured datatype* to model the **ColorDomain**. In this representation, the *datatype fields* hue, saturation, brightness are placeholders for the coordinates of each of the (integral) quality dimensions forming the color domain. In this way the "instances"(members) of ColorDomain are quale vectors $\langle x,y,z\rangle$ where $x \in$ HueValue, $y \in$ SaturationValue and $z \in$ BrightnessValue. The *navigable end name* ***color*** in the association between Apple and ColorValue represents the attribute function ***color*** described above**.**

The two forms of representation exemplified in figures 6.15 and 6.16 do not convey the same information, which we highlight by the use of different stereotypes. In figure 6.15, color objects are one-sidedly existentially dependent on the individuals they are related to via an *inherence* relation. These objects are *bonafide* individuals with a definite numerical identity. As discussed in section 6.2.3, the *characterization* relation between a quality universal and a substantial universal is mapped in the instance level to an inherence relation between the corresponding quality and substantial individuals. In figure 6.16, contrariwise, the members of the ColorDomain are *pure values* that represent points in a quality domain. These values can qualify a number of different objects but they exist independently of them in the sense that a color tuple is a part of quality domain even if no object "has that color".

Both representations are warranted, in the sense that ontologically consistent interpretations can be found in both cases, and which alternative shall be pragmatically more suitable is a matter of empirical investigation. Notwithstanding, we believe that some guidelines could be anticipated. In situations in which the moments of a moment all take their values (qualia) in a single quality domain, the latter alternative (shown in figure 6.16) should be preferred due to its compatibility with the modeling tradition in conceptual modeling and knowledge representation. This is certainly the case with complex quality universals. Additionally, since the conceptualization of these moments depends on the combined appreciation of all their quality dimensions, we claim that they should be mapped in an integral way to a quale vector in the corresponding n-dimensional quality domain.

We can generalize this idea as follows. Let E be an elementary specification capturing the intention of a *substantial or moment* universal U:

$$\forall \mathbf{x}\ (\mathbf{x::U} \rightarrow \exists \mathbf{q}_1,\ldots,\mathbf{q}_n \bigwedge_{i\leq n} (\mathbf{q}_i\mathbf{::Q}_i \wedge i(\mathbf{q}_i,\mathbf{x}))).$$

And a quality specification capturing the intention of each *complex quality universal* universal Q_i in E:

$$\forall \mathbf{q}\ (\mathbf{q::Q}_i \rightarrow \exists \mathbf{k}_1,\ldots,\mathbf{k}_n \bigwedge_{j\leq n} (\mathbf{k}_j\mathbf{::K}_j \wedge i(\mathbf{k}_j,\mathbf{q}))).$$

Let DO_i be the quality domain associated with Q_i. Thus, $DO_i \subseteq D_1 \times_, \ldots \times D_n$ such that *assoc*(K_j,D_j) and *characterization*(Q_i,K_j). Then we can define an attribute function

$$\mathbf{Q_i: Ext(U) \rightarrow DO_i}$$

such that

$$\mathbf{Q_i(x) = \langle d_1,\ldots,d_n\rangle \in DO_i \mid \exists q \exists k_1 \ldots k_n\ (q{::}Q_i) \wedge i(q,x)(\textstyle\bigwedge_{j\leq n} (i(k_j,q) \wedge ql(d_j,k_j)).}$$

Or simply, if we take that (following formula 50) there are functions $\mathbf{K_j: Ext(Q_i) \rightarrow D_j}$ for all $\mathbf{K_j}$ above, then

$$\mathbf{Q_i(x) = \langle d_1,\ldots,d_n\rangle \in DO_i \mid \exists q\ (q{::}Q_i) \wedge i(q,x)(\textstyle\bigwedge_{j\leq n}\ d_j = K_j(q))}$$

Quality domains are composed of integral dimensions. This means that the value of one dimension cannot be represented without representing the values of others. By representing the color quality domain in terms of a structured data type we can reinforce (via its constructor method) that its tuples will always have values for all the integral dimensions. Moreover, the representation of a quality domain should account not only for its quality dimensions but also for the constraints on the relation between them imposed by its structure. To mention another example, consider the Gregorian calendar as a quality domain (composed of the linear quality dimensions days, months and years) in which date qualities can be represented. The value of one dimension constrains the value of the others in a way that, for example, the points [31-April-2004] and [29-February-2003] do not belong to this quality structure. Once more, constraints represented on the constructor method of a data type can be used to restrict the possible tuples that can be instantiated.

In the sequel, we observe the following principle between quality domains and their representation in terms of data types:

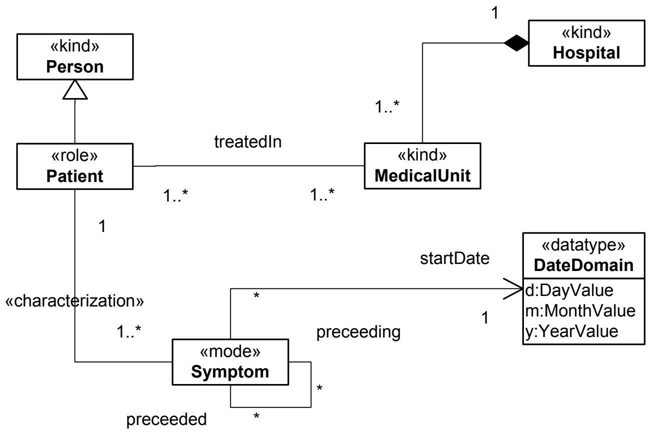
Principle 6.3: Every quality dimension D associated to a quality universal Q may be represented as a datatype DT in a *CM-ontology*; Relations constraining and informing the geometry of a quality dimension D may be represented as operators in the corresponding datatype DT. A collection of integral dimension $D_1 \ldots D_n$ (represented by data types $DT_1 \ldots DT_n$) constituting a quality domain QD can be grouped in structured datatype W representing quality domain QD. In this case, every quality dimension D_i of QD may be represented by a field of W of type DT_i. Moreover, the relations between the dimensions D_i of QD may be represented by constraints relating the fields of data type W.

Principle 6.3 is a generalization of principle 6.2 in order to account for quality domains. In summary, every quality universal Q that is associated to a quality domain in an elementary specification of universal U can be represented in a conceptual model via attribute functions mapping instances of U to quale vectors in the n-dimensional domain associated with Q. The n-dimensional domains should be represented in a conceptual model as an n-valued structured data type.

Now, let us consider a case where one of the moment universals M that characterizes a universal U in its elementary specification is a *mode universal*. We defend here that these are the cases in which we want to explicitly represent a moment universal in a conceptual model. An example of such a situation is depicted in Figure 6.17, which models the relation between a Hospital, its Patients, and a number of symptoms reported by these patients. Suppose an individual patient John is suffering from headache and influenza. John's headache and influenza are moments inhering in John. Even if another patient, for example Paul, has a headache that is qualitatively indistinguishable from that of John, John's headache and Paul's headache are two different individuals. Instances of Symptoms can have moments themselves (such as duration and intensity) and can participate in relations of, for example, causation or precedence.

In figure 6.17, the moment universal Symptom is represented by a class construct decorated with the «mode» stereotype. The formal relation between Symptom and Patient is mapped to the *inherence* relation in the instance level, representing the existential dependence of a Symptom on a Patient. In other words, for an instance *s* of Symptom there must be a specific instance *p* of Patient associated with *s*, and in every situation that *s* exists *p* must exist and the inherence relation between the two must hold. This formal relation has a semantics that is outside the usual interpretation of the association construct in UML. According to its standard usage, the multiplicity 1 in the Patient end only demands that, in every situation, symptom *s* must be related to *an* instance of Patient. The inherence relation, however, requires *s* to be always related to the *one and the same instance* of Patient. The difference between these two sorts of requirements is analogous to those marking the difference between mandatory and essential part-whole relations as discussed in chapter 5, respectively.

Figure 6-17
Representing Quality Universals and Domain Formal Relations



A mode universal such as Symptom in figure 6.17 can be seen as the ontological counterpart of the concept of *Weak entities types* in EER diagrams, which has been lost in the UML unification process. A *weak entity* can be defined as follows (Vigna, 2004):

"a *weak entity* is an entity that exists only if is related to a set of uniquely determined entities, which are called the *owners* of the *weak entity*. For instance, [the] weak entity type edition; each book has several editions, and certainly it is nonsense to speak about an edition if this does not happen in the context of a specific book…When an *entity* is deleted from a *schema instance*, all *owned weak entities* are deleted, too…For entities of type `W` to be owned by entities of type `X`, a requirement must be satisfied: there must be an *identification function* from `W` to `X` that specifies the *owner* of each *entity* of type `W`, that is, a relationship type going from `W` to `X` whose cardinality constraint impose its instances to be functions. Deletion of an entity of type `X` implies deletion of all related entities of `W`…*Weakness* is recursive. If `X` *owns* `Y` and `Y` *owns* `Z`, then `X` *(indirectly) owns* `Z`…as a special case, it must not happen that an *entity owns* itself".

If we perform suitable substitutions to the italicized terms in the above definition, it becomes directly applicable to the notion of moments adopted here: (i) "a *moment* is an entity that exists only if is related to a set of uniquely determined entities, which are called the *bearers* of the *moment"* (formulas 4 and 8 for the inherence relation). Moreover, the requirement that: (ii) when an *object* is deleted from a *conceptual model*, all *inhering moments* are deleted, too" must be satisfied in implementations of the corresponding model, since moments are existentially dependent on their bearers, they cannot exist without the latter. Also the requirement that (iii) "there must be an *identification function* from `W` to `X` that specifies the *bearer* of each *moment* of type `W`, that is, a relationship type going from `W` to `X` whose cardinality constraint impose its instances to be functions", is clearly satisfied by the inherence relation due to the non-migration principle defined for moments (9). Finally: (iv) "*indirect moment of* is recursive. If `X` *inheres* `Y` and `Y` *inheres* `Z`, then `X` *is an indirect moment of* `Z`....as a special case, it must not happen that a *moment inheres in* itself". Again, inherence is irreflexive (5) and although it is intransitive (7), a relation of *indirect moment of* can be defined to be transitive.

To summarize this section we can provide the following procedure to represent in conceptual modeling the elementary specification of universals and their associated quality universals and quality structures:

Take a substantial universal U with its associated elementary specification. For every moment universal Q characterizing U do:

1. If Q is a simple quality universal then principle 6.2 can be applied;
2. If Q is a complex quality universal then principle 6.3 can be applied;
3. If Q is a mode universal then it should be explicitly represented and should be related to U in a model via a *characterization* relation. Moreover, this procedure can be applied again to the elementary specification of Q.

6.3.3 Associations

In most conceptual modeling languages, n-ary relationships are taken to represent sets of n-tuples (e.g., OWL, LINGO, CCT, EER). In UML, the ER concept of a relationship type is called association. *"An association defines a semantic relationship that can occur between typed instances...An instance of an association is called a link...An association declares that there can be links between instances of the associated types. A link is a tuple with one value for each end of the association, where each value is an instance of the type of the end...An association describes a set of tuples whose values refer to typed instances."*(p.81).

The OMG UML Specification is somehow ambiguous in defining associations. An association is primarily considered to be a 'connection', but, in certain cases (whenever it has 'class-like properties'), an association may be a class: An association class is *"[a] model element that has both association and class properties. An AssociationClass can be seen as an association that also has class properties, or as a class that also has association properties. It not only connects a set of classifiers but also defines a set of features that belong to the relationship itself and not to any of the classifiers."*(p.118).

An association A between the classes $C_1, \ldots, C_n$ of a CM-ontology can, in principle, be understood in our framework as a relation (relational universal) *R* between the corresponding universals $U_1, \ldots, U_n$ whose extension consists of all tuples corresponding to the links of A. However, current conceptual modeling languages (including UML) do not distinguish between formal and material relations.

In figure 6.18, an example of a comparative formal relation is the relation of *precedence* between Symptoms. Precedence is a partial order relation between symptoms that depends only on the starting date of each of them.

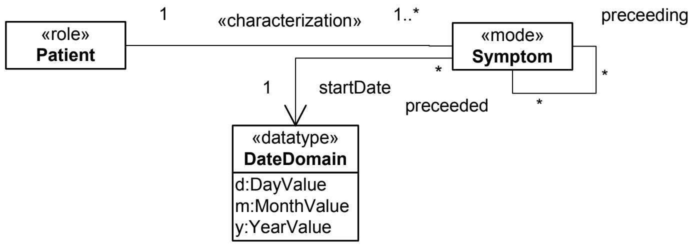


Figure 6-18 Example of a formal Relation between individual modes

It is common in conceptual modeling languages that a number of formal meta-properties (e.g., reflexivity, symmetry, transitivity) are defined for relationships (e.g., LINGO, OWL, RDF, F-LOGIC). In the specific case of *precedence*, these meta-properties are irreflexivity, anti-symmetry and transitivity, i.e., it is a strict *partial ordering* relation. Can we provide an explanation for these meta-properties?

As we have discussed in section 6.2.7, the immediate relata of formal comparative relations are not substantials but qualities. Take, for example, the relations of *taller than*, *heavier than* and *precedence*. All these relations can be reduced to relations between qualities:

- x is *taller than* y iff height(x) > height(y);
- x is *heavier than* y iff weight(x) > weight(y);
- x *preceeds* y iff startDate(x) < startDate(y),

in which *height*, *weight* and *startDate* are attribute functions mapping the substantials x and y to the corresponding qualia. All three quality structures involved in these expressions have a linear structure *ordered* by the <

relation. By making this analysis explicit, it becomes evident that *precedence* is a partial order because the qualities founding this relation are associated with a *total ordered* quality dimension. In general, we can state that the meta-properties of a comparative *formal relation* R_F can be derived from the meta-properties of the relations between qualia associated with the qualities founding this relation R_F. This view is also shared by (Gardenfors, 2000). However, unlike Gardenfors, we do not claim that this holds for all types of relations.

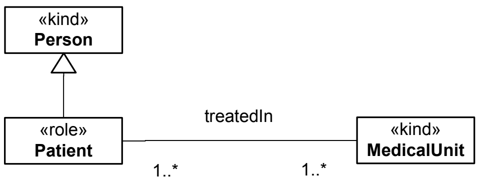
Take for instance the relation treatedIn between Patient and Medical Unit in figure 6.19 below. This relation requires the existence of a third entity, namely an individual Treatment mediating a particular Patient and a particular Medical Unit in order for the relation to hold. This latter case can be modeled in our framework as follows: Let treatedIn be a binary material association induced by a relator universal Treatment whose instances are individual treatments. These individual treatments connect two individuals: a patient, say John, and a MedicalUnit, say TraumaUnit#1. Thus,

$$\mathbf{[John, TraumaUnit\#1]:R_{treatedIn}(Person, MedicalUnit),}$$

since John::Person and TraumaUnit#1::MedicalUnit, and there is a specific treatment t::Treatment such that **(m(t,John) ∧ m(t,TraumaUnit#1)**, We obtain the definition for the tuple [a1,a2] being a link of the association treatedIn between Person and MedicalUnit:

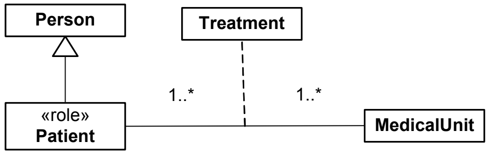
$$\mathbf{[a_1,a_2]:R_{treatedIn}(Person, MedicalUnit) \leftrightarrow a_1::Person \wedge a_2::MedicalUnit \wedge \exists t(t::Treatment \wedge (m(t, a_1) \wedge m(t, a_2)).}$$

Figure 6-19 Example of a Material Relation



How can we represent a material relation in a conceptual modeling language such as UML? Let us follow for now this (tentative) principle: a material relation R_M of the domain may be represented in a CM-ontology by representing the relator universal associated with the relation as an association class. By applying this principle to the treateadIn relation aforementioned we obtain the model of figure 6.20.

Figure 6-20 Representing a Relator Universal as a UML Association Class

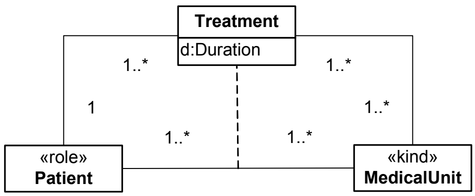


There is a specific practical problem concerning the representation of *material relations* as standard associations that supports a modeling choice in the lines of what is proposed by this principle. This problem, mentioned in (Bock & Odell, 1997a), is caused by the fact that the standard notation collapses two different types of *multiplicity constraints*. For instance, in figure 6.20, the model represents that each Patient can be treated in *one-to-many* Medical Units and that each medical unit can treat *one-to-many* patients. However, this statement is ambiguous since many different interpretations can be given to it, including the following:

- a patient is related to only one treatment to which participate possibly several medical units;
- a patient can be related to several treatments to which only one single medical unit participates;
- a patient can be related to several treatments to which possibly several medical units participate;
- several patients can be related to a treatment to which several medical units participate, and a single patient can be related to several treatments.

The cardinality constraint that indicates how many patients (or medical units) can be related to one instance of Treatment is named *single-tuple* cardinality constraints. *Multiple-tuple* cardinality constraints restrict the number of treatments a patient (or medical unit) can be related to. By modeling the relator universal *Treatment* as an association class one can explicitly represent both types of constraints. A version of figure 6.20 adopting this principle is presented in figure 6.21.

Figure 6-21 Explicit Representation of Single-Tuple and Multiple-Tuple cardinality constraints



This problem is not at all specific to this case. For another example of a situation where this problem arises see figure 6.22. In this case, the (material) relation statement is that: (a) a customer *purchases* one-to-many purchase items from one-to-many suppliers; (b) a supplier supplies one-to-many purchase items to one-to-many customers; (c) a purchase item can be bought by one-to-many customers from one-to-many suppliers.

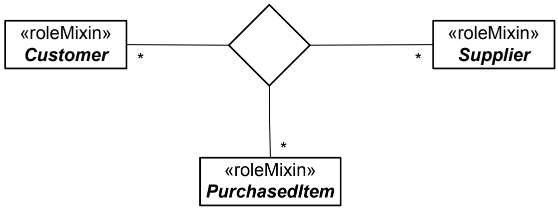


Figure 6-22 Example of a ternary material relation with ambiguous representation of cardinality constraints

PurchaseFrom is a material relation induced by the relator universal *Purchase*, whose instances are individual purchases. Therefore, we have that

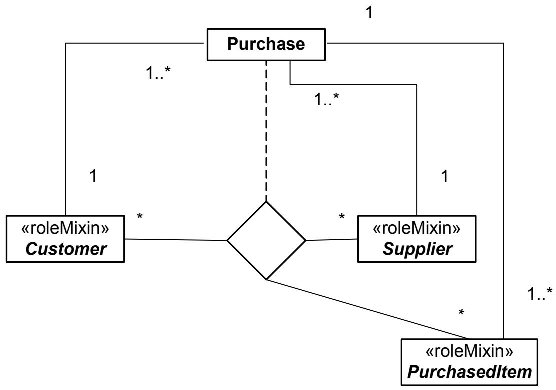
$$\mathbf{[a_1,a_2,a_3]::R_{purchFrom}(Customer,\ PurchaseItem,\ Supplier) \leftrightarrow a_1::Customer \land a_2::PurchaseItem \land a_3::Supplier \land \exists p\ (p::Purchase \land \boldsymbol{m}(p,a_1) \land \boldsymbol{m}(p,\ a_2) \land \boldsymbol{m}(p,\ a_3))}$$

In other words, for this relation to hold between a particular Customer, a particular PurchaseItem, and a particular Supplier, they must be mediated by the same Purchase instance. Once more, we can see that the specification in figure 6.22 collapses single-tuple and multiple-tuple cardinality constraints. For this reason, there several possible ways of interpreting this model, including the following:

- In a given purchase, a Customer participates by buying many items from many Suppliers and a customer can participate in several purchases;
- In a given purchase, many Customers participate by buying many items from many Suppliers, and a customer can participate in only one purchase;
- In given purchase, a Customer participates by buying many items from a Supplier, and a customer can participate in several purchases;
- In given purchase, many Customers participate by buying many items from a Supplier, and a customer can participate in several purchases;

By depicting the Purchase universal explicitly (such as in figure 6.23), we can make explicit the intended interpretation of the material PurchaseFrom relation, namely, that in a given purchase, a Customer buys many items from a Supplier. Both customer and supplier can participate in several purchases. Although a purchase can include several items, each item in this model is a unique exemplar and, hence, can only be sold once.

Figure 6-23 Example of a Material Relation with explicit representation of Single-Tuple and Multiple-Tuple cardinality constraints



This problem is specific to material relations. Formal relations are represented by sets of tuples, i.e., an instance of the relation is itself a tuple with predefined arity. In formal relations, cardinality constraints are always unambiguously interpreted as being *multiple-tuple*, since there is no point in specifying single-tuple cardinality constraints for a relation with predefined arity. Hence, formal relations can be suitably represented as standard UML associations. One should notice that the relations between Patient and Treatment, and Medical Unit and Treatment are formal relations between universals (*mediation*). This is important to block the infinite regress that arises if material relations were required to relate these entities. The same holds for the pairwise associations between Customer, Supplier and PurchaseItem, on one hand, and Purchase on the other.

At first sight, it seems to be satisfactory to represent a material relation by using an association class to model a relator universal that induces this relation. Nonetheless, the interpretation of this construct in the language is quite ambiguous w.r.t. defining what exactly counts as instances of an association class. We claim that the association class construct in UML exemplifies a case of *construct overload* in the technical sense discussed in chapter 2. This is to say that there are two distinct ontological concepts that are represented by this construct. To support this claim, we make use of the following (overloaded) semantic definition of the term as proposed by the pUML community: "an associaton class can have as instances either (a) a n-tuple of entities which classifiers are endpoints of the association; (b) a n+1-tuple containing the entities which classifiers are endpoints of the association plus an instance of the objectified association itself" (Breu et al., 1997).

Take as an illustration the association depicted in figure 6.21. In case (a), *TreatedIn* can be directly interpreted as a *relational universal*, whose instances are pairs *[a,b]*, whereas *a* is patient and *b* is medical unit. In this case, *[a,b]* is an instance of TreatedIn iff there is a relator Treatment connecting *a* and *b*. In interpretation (b), *TreatedIn* is what is named in (Guizzardi & Herre & Wagner, 2002b; Guizzardi & Wagner & Herre, 2004) a *Factual Universal*. In short, if the relator r connects (mediates) the entities $a_1,\ldots,a_n$ then this yields a new individual that is denoted by $\langle r: a_1,\ldots,a_n\rangle$. Individuals of this latter sort are called *material facts*. For every relator universal R there is a set of facts, denoted by *facts*(R), which is defined by the instances of R and their corresponding arguments. We assume the axiom that for every relator universal R there is a factual universal F(R) whose extension equals the set *facts*(R). Therefore, an instance of *TreatedIn* in this case could be the material fact $\langle t_1$: John, $\text{MedUnit}_{\#1}\rangle$, whereas John is a Patient, $\text{MedUnit}_{\#1}$ is a Medical Unit and t_1 is a treatment relator (founded in a treatment process).

The relation between relators, relations and factual universals can be defined as follows. Let R be a relator universal. The factual universal F = F(R) is the basis for the material relation universal Γ(F) whose instances are n-tuples of entities. In general, a relation universal Γ(F) can thus be defined by the following schema. Let $\phi(a_1,\ldots,a_n)$ denote a condition on the individuals $a_1,\ldots,a_n$

$$[a1\ldots an]::\Gamma(F)\ (U1\ldots Un) \leftrightarrow \bigwedge_{i\leq n} ai::Ui \wedge \phi(a1\ldots an)$$

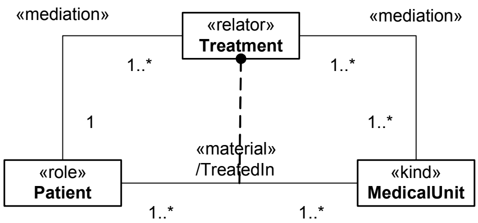
A relation is called *material* if there is a relator universal *R* such that the condition ϕ is obtained from R as follows: $\boldsymbol{\phi(a_1\ldots a_n) \leftrightarrow \exists k\ (k::R \wedge \langle k:a_1\ldots a_n\rangle::F(R))}$.

As a moment, a relator can bear other moments. For example, in figure 6.21, the temporal duration of a Treatment is a moment of the latter. For this reason, between the two aforementioned interpretations for association classes, we claim that interpretation (b) should be favored, since it allows for the explicit representation of relators and their properties. However, there is still one problem with this representation in UML. Suppose that treatment t_1 mediates the individuals John, and the medical units $\text{MedUnit}_{\#1}$ and $\text{MedUnit}_{\#2}$. In this case, we have as instances of Treatment both facts $\langle t_1$: John, $\text{MedUnit}_{\#1}\rangle$ and $\langle t_1$: John, $\text{MedUnit}_{\#2}\rangle$. However, this cannot be represented in such a manner in UML. In UML, t_1 is supposed to function as an object identifier for a unique tuple. Thus, if the fact $\langle t_1$: John, $\text{MedUnit}_{\#1}\rangle$ holds then $\langle t_1$: John, $\text{MedUnit}_{\#2}\rangle$ does not, or alternatively, John and $\text{MedUnit}_{\#2}$ must be mediated by another relator. These are, nonetheless, unsatisfactory solutions, since it is the very same relator Treatment that connects one patient to a number of different medical units.

We therefore propose to represent relator universals explicitly as in figure 6.24. This model explicitly distinguishes the two entities: relator universals are represented by the stereotyped class «relator»; material relations are represented by a derived UML association stereotyped as «material». The dashed line between a material relation and a relator universal represents that the former is derived from the latter (see *derived from* relation figure 6.13). To mark this difference to the similar graphic symbol in UML, we attach a black circle in the relator universal end of this relation. In this figure, a particular Treatment is existentially dependent on a single Patient and in a (immutable) group of medical units. This would mean in UML that for every association representing an existential dependency relation between a moment and the endurant(s) it depends on, the association end should be frozen in the side of the latter. This compound modeling construct should replace the ambiguous association class construct in UML.

Unlike in figure 6.21, the entities representing a relator universal (the stereotyped class that takes the place of an association class), and the material relation (the association itself) are distinct entities. In fact, the latter is completely derived from the former (see definition 6.16). For instance, the relator universal Treatment and the material relation TreatedIn represent distinct entities and can possibly have different cardinalities, since the same relator t_1 can connect both the entities in [John, $MedUnit_{\#1}$] and [John, $MedUnit_{\#2}$]. Nonetheless, the cardinality constraints of TreatedIn can be completely deduced from the existential dependency relations between Treatment and the universals whose instances are the relata of TreatedIn, namely, Patient and MedicalUnit.

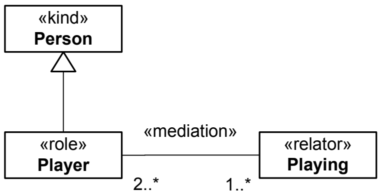
Figure 6-24
Specification with explicit representation of a Relator Universal, a Material Relation, a (formal) derivation relation between the two



This representation eliminates the construct overload present in figure 6.21, since all constructs here have a unique and unambiguous ontological interpretation. Although (derived) relations can be important representation constructs, serving as a base for the specification of *mappings* in design time (Bock & Odell, 1997b), it is important to recognize the conceptual and ontological primacy of their foundations. Comparative formal relations are completely founded on certain intrinsic moments, and material relations are founded on relators. Moreover, as discussed in depth in (Snoeck & Dedene, 1998), the explicit representation of relator universals and their corresponding existential dependency relations provides a suitable mechanism for consistency preservation between static and dynamic conceptual models.

Another benefit of this approach is that relator universals allow for the representation of anadic relations, i.e., relations whose arities are not fixed. Take for example the relation between some people that *play together*. For each instance of *playing* there may be a different number of people that participate in it (Loebe, 2003). This situation can modeled in the approach adopted here by explicitly representing of the anadic relator universal *Playing*.

Figure 6-25 Using Relator Universals to represent Anadic Relations



In figure 6.25, the relator *Playing* connects individuals that play the same role. In (Loebe, 2003), this is named *intensional symmetry* (in contrast to the more common extensional symmetry that holds for formal relations). In general, a material relation R_M is intensionally symmetric iff the relator universal U_R from which R_M is derived is such that: every instance of U_R mediates only entities that instantiate the same role universal. It is important to emphasize that both comparative relations and material relations, as derived entities, have their meta-properties also derived from their foundations. In the case of formal relations, they are derived from the meta-properties of relations among qualia in the underlying conceptual space. In the case of material relations, they are derived from their founding relators and mediated entities. According to formula (45), relators must connect distinct entities. As a consequence, it should be clear that, there cannot be intensionally reflexive binary material relations.

The benefits of this approach are even more evident in the case of n-ary relations with n > 2. Take the UML representation of a ternary relation in figure 6.22. In this specification, we are forced to represent the minimum cardinality of zero for all association ends. As explained in the UML specification (p.82): *"For n-ary associations, the lower multiplicity of an end is typically 0. If the lower multiplicity for an end of an n-ary association of 1 (or more) implies that one link (or more) must exist for every possible combination of values for the other ends"*. As recognized by the UML specification itself, n-ary associations in which there are tuples for every possible combination of the cross-product of the extension of the involved classes are atypically. Thus, in the majority of cases, the UML notation for n-ary associations completely looses the ability of representing real minimum cardinality constraints. Furthermore, as empirically demonstrated in (Bodard et al., 2001), conceptual models without optional properties (minimum cardinality constraints of zero) lead to better performance in problem-solving tasks that require a deeper-level understanding of the represented domain.

Finally, in the same way as qualities, relators can have their own inhering moments (e.g., Duration, as a quality associated to the universal Treatment, in figure 6.21) but also be mediated by other relators such as, for instance, a relator universal *Payment* whose instances connect particular Treatments and Payers.

The results of this section can be summarized in the following principle regarding the representation of formal and material relations in a *CM-ontology*:

Principle 6.4: In a *CM-ontology*, any formal relation universal R_F of the domain may be directly represented as a standard association whose links represent the tuples in the extension of R_F. Conversely, a material relation R_M of the domain may be represented in a *CM-ontology* by a complex construct composed of: (i) *CM-class* stereotyped as «relator» representing the *relator universal*. The relator universal is associated to CM-Classes representing mediated entities via associations stereotyped as «mediation»; (ii) a standard association stereotyped as «material» representing a material relation whose links represent the tuples in the extension of R_M; (iii) a dashed line with a black circle in one of the ends representing the formal relation of derivation between (i) and (ii), in which the black circle lies in the association end of the relator universal.

6.4 Related Work

The approach found in the literature that is closest to the one presented here is the so-called BWW approach presented in (Shanks & Tansley & Weber, 2003; Evermann and Wand, 2001a,b; Wand & Storey & Weber, 1999; Weber, 1997; Wand & Weber, 1995, 1993, 1990, 1989). In these articles, the authors report their results in mapping common constructs of conceptual modeling to an upper level ontology. Their approach is based on the BWW ontology, a framework created by Wand and Weber on the basis of the original metaphysical theory developed by Mario Bunge in (Bunge, 1977, 1979).

In this section we compare the foundation ontology proposed here with BWW in terms of their theories and of their corresponding mapping approaches. Additionally, we discuss the approach proposed in (Veres & Hitchman, 2002; Veres & Mansson, 2005). This approach applies a psychologically and linguistically well-founded ontology as well as empirical evidence to criticize the general assumptions of the BWW approach, and in particular, its proposals for the representation of relational properties.

6.4.1 Things and Substantials

The concepts of *substantial* here and of *thing* in BWW are both based on the Aristotelian idea of substantial, i.e.,

1. an essence which makes a thing what it is;
2. that which remains the same through changes;
3. that which can exist by itself, i.e., which does not need a 'subject' in order to exist.

In BWW, a thing is defined as a substantial individual with all its substantial properties: "a thing is what is the totality of its substantial properties" (Bunge, 1977, p.111). As a consequence, in BWW, there are no bare individuals, i.e., things without properties: a thing has one or more substantial properties, even if we, as cognitive subjects, are not or cannot be aware of them. Humans get in contact with the properties of things exclusively via the thing's attributes, i.e. via a chosen representational view of its properties. This is far from saying that Bunge himself embraces the so-called *Bundle of Universals* perspective. In fact, he explicitly rejects this theory and, instead, holds a position that can be better identified with a *substance-attribute* view (Armstrong, 1989).

In short, in the former type of theory, particulars are taken as *bundles of universals*, i.e., as aggregates of properties which themselves are repeatable abstract entities. The most important exemplar of this type of theory was proposed by Russel in (Russel, 1948). A direct consequence of this theory is that two different things cannot have the same properties, where properties are universals. It is important not to confound this principle with another principle discussed in chapter 4 named the Leibniz's law, also know as the principle of the (P1) *indiscernability of identicals*. Principle (P1) states that if x is identical to y then whatever property x has then y has as well, or formally

$$\forall x,y\ (x = y) \rightarrow \forall \mathbf{U}\ (x::\mathbf{U} \leftrightarrow y::\mathbf{U})$$

The principle is implied by the bundle theory of universals is named the (P2) *Identity of indiscernibles* and can be formally stated as follows

$$\forall \mathbf{x},\mathbf{y} \forall \mathbf{U}\ (\mathbf{x}::\mathbf{U} \leftrightarrow \mathbf{y}::\mathbf{U}) \rightarrow (\mathbf{x} = \mathbf{y})$$

that is, if two individuals instantiate the same universals then they are identical. Whilst principle (P1) is universally accepted among philosophers, (P2) is matter of great controversy, and there is at least the logical possibility that (P2) fails to be the case (Armstrong, 1989).

Another problem with this bundle theory is that it makes universals the substance of reality, in the sense that everything is constructed out of universals. How can be the case that the concrete reality is made solely by these abstract entities? These problems (among others) are discussed in great detail in (Armstrong, 1989, chap.4). In fact, among the theories that countenance the existence of properties, Armstrong considers the bundle theory of universals to be the weakest from a philosophical point of view.

The *substance-attribute* view makes an explicit distinction between a thing and the properties that the thing happens to have. As a consequence, the theory countenances the existence for every individual of a propertyless *substratum, particularized essence* or *bare particular*. The notion of substratum is strongly associated with the British empiricist philosopher John Locke (Armstrong, 1989) and due to its mysterious nature it has been the target of some criticism throughout history. Nonetheless, as a "theoretical fiction" (Bunge, 1977, p.57) it solves some of the philosophical problems existing in the *bundle of universals* theories.

In summary, Bunge is a *realist w.r.t. universals*, i.e., he claims that universals exist in reality independent of our knowledge of their existence. However, he denies the existence of *particularized properties*, i.e., moments. The denial of property instances puts BWW in a singular position among the foundational ontologies developed in the realm of computer science (e.g., Schneider, 2003b, 2002; Heller et al., 2004; Masolo et al., 2003a; Neuhaus & Grenon & Smith, 2004).

In principle, it seems that a thing in BWW could be directly associated to our concept of substance. However, there are some important differences between the two. Whilst a BWW-thing can be thought as a substratum instantiating a number of properties (as repeatable abstract entities), our substantials are particulars that bear other particularized properties (i.e., moments), or to borrow Simons' phrase, "particulars in particular clothing" (Simons, 1994).

Although we do not make any ontological commitment w.r.t. the nature of our *substratum*, by adopting a four-category ontology, if necessary, we can dispense with a substratum of a mysterious nature. In this case, we can take a view such as the one of Simons' *Nuclear Theory*. This theory proposes that the accidental properties instances of an individual are held together by their mutual existential dependency to a nucleus. This nucleus, in turn, is composed by a number of essential properties that form what Husserl names a *foundational set*, i.e., a closure system under the relation of existential dependency. This approach has the advantages of the substance-attribute view, without having to accept its problems, since the nucleus is akin to a substratum, only not a mysterious one. In BWW, contrariwise, the mysterious substratum cannot be eliminated without putting the theory into a *Bundle of Universals* group. We claim that this flexibility puts an ontology in which moments are recognized in a better position than one in which they are not.

6.4.2 Properties and Moments

Despite the differences, an important commonality between a BWW-thing and our notion of substance is that, as the former, the latter has a non-empty appearance, i.e. every substance bears at least one moment. The converse is also true, i.e., in both approaches a property exists only in connection with its bearer. In BWW, a property whose existence depends only on a single thing is called an *intrinsic property*. A property that depends on two or more things is called a *mutual property*. These concepts are analogous to our notions of *intrinsic* and *relational moment universals*. Nevertheless, once more, in our approach properties are instantiated. Thus, our intrinsic properties can be better defined as universals whose instances inhere in a single individual, while relational properties are universals whose instances mediate multiple individuals.

In BWW, only things possess properties. As a consequence, a property cannot have properties. This dictum leads to the following modeling principle: "Associations should not be modeled as classes" (Rule 7 in Evermann & Wand, 2001b). Here, in contrast, according to principle 6.1 we propose that any first-order universal can be represented as a conceptual modeling class. This issue marks an important divergence between the view proposed here and the BWW approach.

The BWW authors claim that classes in a conceptual model of a domain should only be used to represent substantial universals. In particular, they deny that universals whose instances are particularized properties (i.e., moments) should be represented as classes. This claim is not only perceived as counterintuitive by conceptual modeling practitioners (as shown by Veres & Hitchman, 2002; Hitchman, 2003; Veres & Mansson, 2005), but it is also controversial from a metaphysical point of view.

Bunge denies that there properties of properties, i.e., higher-order properties. Since classes can have attributes representing properties, (Wand & Storey & Weber, 1999; Weber, 1997) claim that properties should not be represented as classes, since in this case they could have attributes ascribed to them. We think there are some problems with this argumentation.

First, claiming that there are no higher-order universals is itself quite controversial. (Armstrong, 1989), for instance, who embraces scientific realism as a theory of universals, claims that higher-order properties are *necessary* to represent the concept of a *law*. For Armstrong, a law such as Newton's F = MA describes a second-order relation between the three universals involved. Strangely enough, Bunge also defines the concept of a *Law* (quite a central notion is his approach) as a relation between properties, which then makes it a second-order relation (Bunge, 1977, p.77). The view that there are, in fact, material higher-order universals is also shared by other approaches (e.g., Degen et al., 2001; Heller & Herre, 2004). Even simple higher-order relations between universals such as "*Redness is more like Orange than it is like yellow"* cannot be dealt with in the current version of the BWW framework. As discussed in section 6.2.6, in the approach presented here, if one wants to dispense with higher-order properties, this relation can be expressed in terms of relations between quality regions (abstract individuals) in a conceptual space.

The second problem with Wand & Weber's argumentation is that, even if one denies the existence of higher-order properties, it is not necessary to proscribe the representation of properties as classes. Alternative, one can simply proscribe the representation of attributes in classes representing properties[58].

Nonetheless, if one subscribes to Bunge's theory, there is a much stronger reason to argue against the representation of non-substantial universals as classes. Since Bunge denies the existence of *particularized properties*, one could simply state that properties should not be represented as classes because they should not be allowed to have instances. As discussed in section 6.2.3, there is philosophical, cognitive and linguistic support in the literature for taking an ontological view of individuals in which moments are accepted (Schneider, 2003b, 2002; Heller et al., 2004; Masolo et al., 2003a; Neuhaus & Grenon & Smith, 2004; Lowe, 2001). Moreover, even if both ontological choices were deemed equivalent, there are cognitive and pragmatic reasons for defending the acceptance of property instances and, hence, in favor of accepting also the representation of non-substantial universals as conceptual modeling types. As demonstrated in section 6.3.3, the explicit representation of relator universals (relational properties) allows for the explicit representation of

[58]Since the authors take attributes to represent substantial properties, we assume here that the restriction is meant only for this type of attributes. That merely formal higher-order attributes (predicates) can always be created for universals is beyond dispute.

single-tuple and multiple-tuple cardinality constraints in associations. Additionally, as discussed in the same section, it enables the possibility of consistency preservation between static and dynamic conceptual models of a domain. Finally, in our approach, properties of properties such as the *hue of a certain color* or the *graveness of a certain symptom* can be modeled as first-order inherence relations between moments, and this is possible exactly because we countenance particularized properties.

To provide one more example of the importance of relators in conceptual modeling, suppose the situation in which one wants to model that *students* are *enrolled* in *universities* starting in a certain date. Following the proscription of mutual properties being modeled as entity types, (Wand, Storey, and Weber 1999) propose the following model for this situation (figure 6.26)[59].

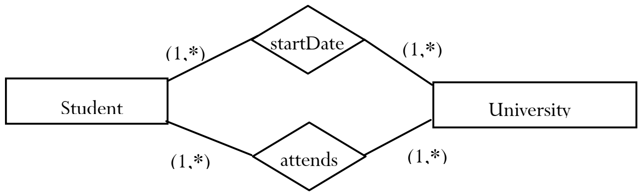


Figure 6-26 An alternative modelling of "properties of properties" according to the BWW approach (from Wand & Storey & Weber, 1999).

We claim that it is rather counterintuitive to think about a model of this situation in these terms. According to Wand, Storey & Weber, relationships representing mutual properties are equivalent to n-ary attribute predicates. In this case, what is startDate supposed to stand for? Is it a binary predicate that holds, for example, for John and the University of Twente, like in *startDate (John, UT)*? This seems to be an absurd conclusion. Thus, startDate should at least be a ternary predicate applied to, for instance, *startDate (John, UT, 14-2-2004).* Now, suppose that there are many predicates like this one relating a student and a university. For example, the *start-date of writing the thesis*, the *start-date of receiving a research grant*, etc. We believe that, in this case, the authors would propose to differentiate the startDate depicted in figure 6.26 by naming it *startDateofEnrollment*. But does not this move make it obvious that *startDate* is actually a property the enrollment? In our approach, this can be explicitly modeled such as in figure 6.26.

[59]The BWW model is shown in the ER notion according to the original models presented in (Wand, Storey, and Weber 1999).

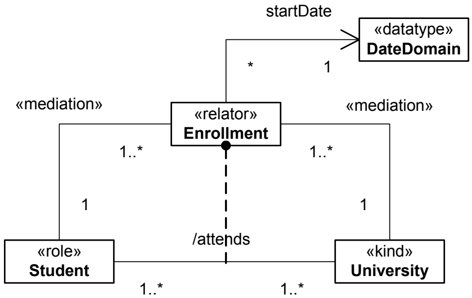


Figure 6-27 The representation of "properties of properties" according to our approach

The model of figure 6.27 makes an explicit distinction between a closed-linked relation between student and university and an indirect relation between student and start date. In what follows we discuss cognitive and linguistic justifications for highlighting this distinction.

6.4.3 Conceptual Structures and a Linguistic Analysis of n-ary relationships

In a series of papers, Veres and colleagues (Veres & Hitchman, 2002; Veres & Mansson, 2005; Hitchman, 2003) offer a detailed analysis and criticism of the general assumptions of the BWW approach. More specifically, in (Veres & Mansson, 2005), they provide empirical evidence to support a case against the BWW treatment of associations.

They argued that claims about the intuitiveness of modeling languages constructed by the principles provided by the BWW approach are based on a hidden assumption about psychology. That is, "the assumption behind the claim that an 'ontologically correct' modeling language will be intuitive is that people's cognition is also 'ontologically correct' at some level: if our models represent reality the way it 'really is', then people will find this a natural representation" (Veres & Mansson, 2005). The criticism cannot be against ontology *per se*, since the authors themselves state that they "describe an ontology of conceptual structure" or "psychologically motivated ontology" for the same purpose, but against the use (for this purpose) of a revisionary ontology (such as Bunge's) that lacks a linguistic and cognitive foundation. Moreover, they observe that the human cognitive architecture imposes fundamental constraints on our representations and that *some* prescriptions on the representation of entities should also be based on psychology

In contrast, the authors propose the use of a psychologically motivated ontology based on the theory of conceptual structures developed by the linguist Ray Jackendoff (Jackendoff 1983, 1990, 1997) as a foundation for guiding conceptual modeling decisions. It is important to emphasize that conceptual structures themselves are not language dependent, and they are not determined by language use. Quite the contrary; language evolved as a way to externalize the pre existing content of cognition for the purpose of communication (Pinker & Bloom, 1990). The idea is that we gain insight into concepts by observing the correspondences between the structural properties of languages, on one side, and conceptual structures on another. This idea is very much conformant with approach developed here (see discussions on chapters 2 to 4 of this thesis).

In (Veres & Mansson, 2005), the authors propose an interesting example that considers the linguistic (syntactic) distinction between *complements* and *adjuncts*. Observe the following sentences:

1. Sarah robs the 7-11 in New York.
2. Adam robs in Washington in February.

The basic premise in both sentences is that someone commits a robbery. However, if one constructs a syntactic tree for these sentences it becomes manifest that *robs* and *7-11* in sentence 1 are more closely linked than are *robs* and *Washington* in sentence 2. *7-11* is a *complement* of the verb *robs*, whereas *in Washington* is a less closely linked *adjunct*. Simple transformations on these sentences can show that a verb is quite selective in the complements it will take, but insensitive to its adjuncts. An interesting point observed by the authors is that this distinction reflects a distinction in the underlying conceptual structure: following a lexical/conceptual correspondence rule from (Jackendoff, 1990) it can be shown that the complement is directly linked to the verb in the conceptual structure. Adjuncts, conversely, are mapped into some lower position in the recursive definition.

Now, suppose a conceptual model containing the entities *Person, Location, Establishment, Time_period.* How should the situations in (1) and (2) be modeled? Both situations describe a ternary relationship between three entities: *person-establishment-location* or *person-location-time_period*. And, as (Wand, Storey, and Weber 1999) suggest, all n-ary relationships should be modeled alike. However, as Veres and colleagues demonstrate, the two scenarios have subtly different underlying conceptual representations.

Figure 6.28 and 6.29 depicts two alternative models of situations (1) according to the BWW approach and to ours, respectively. First of all, the use of a relator universal in our model eliminates the ambiguity caused by the collapse of multiple-tuple and single-tuple cardinality constraints in 6.28. Additionally, since it is not the case that there are tuples for all the combinations of instances of Thief, Establishment and Location, in figure 6.28, the minimum cardinality constraints for all association ends connected to these classes must be specified equal to zero. This problem disappears in figure 6.29.

Notice that the specification of figure 6.28 contradicts the BWW rule that proscribes the representation of optional properties (Weber, 1997). In other words, the BWW model represented in this figure is inconsistent even w.r.t. to the rules prescribed by the BWW approach. Moreover, this will be the case for almost all n-ary relations with n > 2.

Finally, the approach taken in figure 6.29 allows for the explicit representation of the relations between Thief (Person) and Establishment (a material relation mediated by a relator robbery) and the indirect relation between Thief and Location. Notice that *robs* can be hardly said to be a real ternary relation in the first place, since Location and Establishment are strongly coupled. In fact, once the establishment is fixed the location of the robbery is derived from the location of the establishment.

Figure 6-28 The representation of the sentence *Person robs Establishment in Location* according to the BWW approach

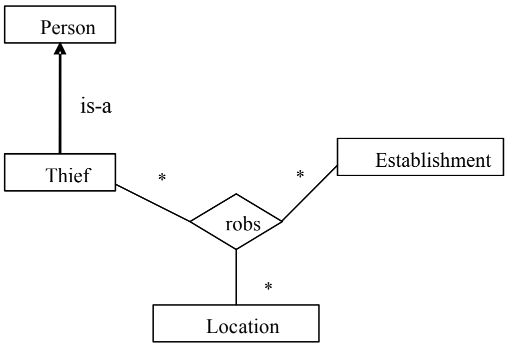
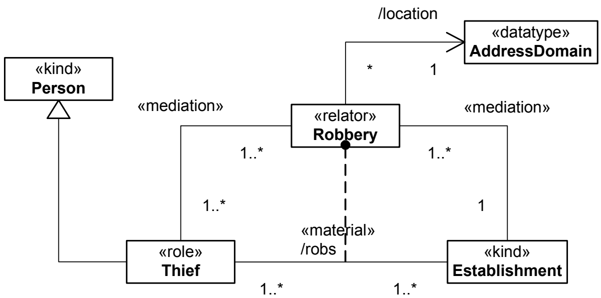


Figure 6-29 The representation of the sentence *Person robs Establishment in Location* according to our approach



Despite what has been discussed so far, the claim that "the practice of representing domain attribute and relationship types as UML classes is *ontologically incorrect*" made by the BWW authors is, in our view, not-warranted. At maximum, one can say that this practice produces conceptual models that are non-conformant with the BWW ontology. However, models obeying this practice would conform to several other philosophically sound foundational ontologies in computer science (e.g., Heller et al., 2004; Masolo et al., 2003; Schneider, 2002; Neuhaus & Grenon & Smith, 2004, Veres & Mansson, 2005). In fact, we believe that the evidence shown here should support not only a case in favor of our ontological choices, but also a case against the suitability of Bunge's ontology as a proper ontological foundation for conceptual modeling.

6.4.4 Natural Kinds and Substantial Universals

In BWW, the definition of a class is based on the notion of the scope of a property. A scope s of a property P is a function assigning to each property that exists in a domain a set of things from that domain, i.e., s(P) is the set of things in the domain that possess property P. A class is then defined as the scope of a property.

If we have a non-empty set P of properties, the intersection of the scopes of all members of P is called a kind. Finally, a kind whose properties satisfy certain *laws* is called a *natural kind*.

A model that describes things with common properties is named a *functional schema.* A functional schema comprises a finite sequence of functions $F = \langle F_1..F_n \rangle$, such that each function F_i (named an *attribute*) represents a property shared by the members of the class described by the functional schema. For every attribute F_i there is a co-domain V_i of values. Evermann and Wand claim that a CM-type cannot be mapped to any of the BWW concepts of class, kind or natural kind, because the latter are defined extensionally (as special sets), while the former is defined intensionally and has an extension at run-time. They, therefore, propose that a CM-type is equivalent to a functional schema of a natural kind.

As we discuss in section 4.5.1, the notion of a natural kind is equivalent to our *substance sortal*, or simply, a *kind*. In that section, we argue that a kind is only one of the many types of substantial universals that are needed for conceptual modeling. Clearly, some of the classifiers that are used in the examples provided by the BWW authors (e.g., student in figure 6.26) are not natural kinds according to this definition.

Evermann and Wand propose that the conceptual modeling representations of Natural Kinds (i.e., functional schemas) should be accompanied of a constraints specification that restricts the possible values that the attributes of a substantial thing can assume. This idea has its origins in Bunge's ontology itself. Bunge defines a function F(t) as the *state function* of the thing, such that $F(t') = \langle F_1(t')..F_n(t') \rangle$ is said to represent the *state of a* *thing* at time t'. The set $V_1 \times \ldots \times V_n$ (i.e., the Cartesian product of all co-domains) is termed the *state space of a thing*. Now, since the properties of instances of a natural kind are lawfully related, it is not the case that the coordinates of state vectors can vary freely. The subset of $V_1 \times \ldots \times V_n$ constrained by the laws of the natural kind being described is named by Bunge the *lawful state space* of a thing. In other words, the lawful state space associated with a natural kind defines all possible states that instances of this kind can assume.

In our approach the *lawful state spaces* are the realist counterpart of conceptual spaces associated with *substance sortals*. As in the BWW, we claim that the conceptual model representation a universal U should be constrained by the laws relating the properties of U, or in a less than ideal case, by the known structure of the conceptual space associated with U. Nonetheless, unlike in BWW, we acknowledge that the value domains V_i can themselves be multidimensional, exhibiting a constrained structure that occurs in the definition of quality domains associated with possibly different substantial universals. For this reason, we believe the explicit representation of quality domains (associated with quality universals) as datatypes not only provides a further degree of structuring on lawful state spaces, but it also allows for a potential reuse of specifications of a subset of its constraints.

6.5 Final Considerations

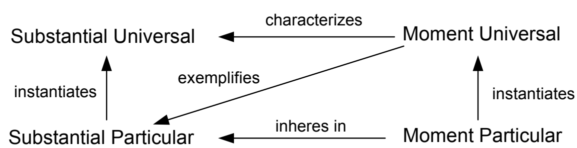
In this chapter, we provide ontological foundations for some of the most basic conceptual modeling constructs, namely, classes, attributes and relationships. These are fundamental constructs in the sense that they are present in practically all conceptual modeling languages (e.g., UML, ER, LINGO, CCT), and in particular, in the so-called *semantic web languages*. However, in these languages, these constructs are understood only on a superficial level, and typically interpreted as sets of elements, or as the extensions of certain modeling *predicates*. According to (Wand & Storey & Weber, 1997) the relationship construct seems to be particularly problematic. Despite of being perceived as very important in practice and widespread in the literature, empirical evidence shows that the use of this construct is often problematical as a way of communicating meaning in an application domain. For example, in studying the process of logical database design, (Batra & Hoffler & Bostrom, 1990, p.137) conclude "that the most commonly occurring errors pertain to the connectivity of relationships."

In pace with Wand and colleagues, we believe that this is mainly due to the lack of consensus and imprecise definitions of the meaning of these categories. As a consequence, "users of conceptual modeling languages are frequently confused about whether to show an association between things via a relationship, an entity, or an attribute." (Wand & Storey & Weber, 1999, p.494). In fact, an inspection in the conceptual modeling literature shows that these constructs receive a mere superficial characterization in most conceptual modeling languages, such as that "associations represent a semantic link between objects" (OMG, 2003c). Thus, in practice, a relationship is simply taken to stand for any n-ary predicate in natural language.

In this chapter, we have shown that taking entity and relationship universals as sets does not provide any explanation for what it means for something to have a certain property (e.g., a certain atomic number), or for multiple things to be related (e.g., John being married to Mary). For this reason, any philosophically and cognitively well-founded ontology must countenance a number of more fundamental urelements that should complement the set-theoretical ontologies underlying existing conceptual modeling languages. To put it in a different way, *Barrichello's Ferrari is red* because it bears its particular redness, and *Peter is kissing Sarah* because there is a particular kiss in which both of them participate.

The foundational ontology developed in this chapter is centred in an Aristotelian ontological square depicted in figure 6.30. Ordinary mesoscopic objects of our every-day experience belong to the category of substantial. Examples are an apple, a cat, my car, Queen Beatrix, the Dutch part of the North Sea, etc. These objects are characterized by certain particular qualities. For instance, an apple *x* has a certain particular color *y*, which is a quality of this apple, and numerically distinct from the particular colors of all other apples. Because *x* bears y, which is an instance of a universal Y, *x* is said to exemplify Y, in the way, for example, that Monica Bellucci is said to exemplify beauty, or that Stephen Hawking is said to exemplify intelligence. Additionally, if every x exemplifies a number of (moment) universals $Y_1 \ldots Y_n$, we say that these universals characterize the (substantial) universal X that x instantiates[60].

Figure 6-30 The Ontological Square



As we have discussed along the chapter, there is strong support in the cognitive and philosophical literature for accepting the existence of particularized properties and particularized relations. Actually, in agreement

[60]This is far from saying that the *meaning* of a substantial universals correspond to an enumeration of properties common to all its instances (see discussion in Keil, 1992).

with (Schneider, 2002), we believe that moments are indeed "the immediate objects of everyday experience". From an ontological point of view, the acceptance of these entities enables the construction of a foundational ontology in which minimum metaphysical commitments are made. In section 6.4.1, we have discussed that it allows for an ontology that can dispense both with the controversial "identity of the indiscernibles" principle, and with a mysterious substratum or bare particular. However, another philosophical advantage can be pointed out in favour of this approach. In section 6.2.5, we are intentionally neutral w.r.t. to the nature of universals. However, in an ontology that accepts moments, there is open the possibility of conceiving universals as special sorts of particulars, such as, for example, *resemblance structures* (Armstrong, 1989; Schneider, 2003b). The advantage of such a view is that, if being a first-order particular is to be part of a resemblance structure then the same can be said for higher-order ones. In fact, in this case, higher-order instantiation could be explained in terms of mereological relations between resemblance structures. For instance, if the first-order universal *Eagle* is thought of as a particular then not only it can bear its own moments (e.g., *life expectancy*), which are not exemplified in any particular eagle, but it easily can be thought of as a instance (part of) the higher-order universal (resemblance structure) *Bird species*. There is support in the philosophy of biology literature for conceptualizing biological species as individuals, or more precisely as integral wholes unified by the characterizing relation of common ancestry (Erenefsky, 2004; Milikan, 1998). Although we shall not entertain this possibility here, it should be investigated in future extensions of our ontological framework. This idea, in principle, could be used to provide an ontological foundation for the important (but only vaguely explained) concept of *Powertype* in conceptual modeling.

One formal relation that plays a paramount role here is *existential dependency*. This simple notion, which has been formally defined in chapter 5 allow us to precisely characterize: (i) the distinction between substantial and moments; (ii) inherence (mediation) relation between moments and substantials. In other words, once we have a formal notion of existential dependence, we can differentiate which particulars in the domain are *substantials* and which are *moments*. Based on the multiplicity of entities a particular depends on, we can distinguish between *intrinsic* and *relational moments*. With qualities we can explain *comparative formal relations*, and with relators, the *material* ones. In particular, by employing (explicitly represented) relators, we can provide not only an ontologically well-founded interpretation for the (otherwise problematic) relationship construct, but also one that can accommodate more subtle linguistic distinctions. Additionally, from a conceptual point of view, we can produce models that are free of cardinality constraint ambiguities. Furthermore, with the notions of modes and relators we can define *qua individuals*. Finally, with moments we can explain exemplification, and with exemplification characterization. In summary, practically the whole core of the urelement segment of our foundational ontology is based on the primitive and unambiguous notion of *existential dependency*.

In this chapter, we have also been intentionally neutral to whether substantials and moments should be interpreted as *continuants* or as *snapshot entities* (i.e., momentary states of continuants)[61]. In the latter case, moment continuants can be thought as logical constructions from world-bounded moment snapshots, in the same way that in L_1 (see section 4.4.1) substantials are considered as logical constructions from world-bounded substantial snapshots. We name these moment counterparts of substantial individual concepts as *moment persistents* (after Heller et al., 2004). This interpretation can provide an interesting explanation for some linguistic cases regarding properties. Take the following example from (Masolo et al., 2003a):

1. This rose is red.
2. Red is a color.
3. This rose has a color.
4. The color of this rose turned to brown in one week.
5. The rose's color is changing.
6. Red is opposite to green and close to brown.

In (1), the predicate red is applied to an individual rose. As an adjective, red is a characterizing universal of the substantial universal rose, i.e., a mixin. In (2), red is intended as a noun, i.e., red is taken as abstract particular, a quality region in a quality domain (another abstract particular). The same hold for (6), in which a particular shade on red can be interpreted as a region of a quality domain that, due to the structure of that domain, bears certain relations to some other regions (brown and green). Sentence (3) can be interpreted as a case of inherence. Specially, when taken together with (4) and (5). In sentences (4) and (5) one is not speaking of a shade of color, neither of a characterizing universals but of something that changes while maintaining its identity. In DOLCE, this is explained by having a quality color that can change with time. This position is at odds with most approaches in the literature that take even super-determinate moment universals to be rigid. In other words, if a particular is an instance of redness, it cannot cease to be so, ergo, it cannot become and instance of brownness. Therefore, in most theories of particularized

[61] See discussion in section 4.4.

properties, these changes are explained by having one particular quality replaced by another. However, in this case, there is no quality there that persists while keeping its identity. Now, if individuals are interpreted as snapshots then we can have:

(i) in world w_1 a moment x_1 instance of color c_1 inhering in r_1 (rose snapshot);

(ii) in world w_2 a moment x_2 instance of color c_2 inhering in r_2 (rose snapshot);

(iii) r_1 and r_2 are states of the same continuant rose because there is an individual concept R (in the extensional of the universal rose) such that $R(w_1) = r_1$ and $R(w_2) = r_2$;

(iv) x_1 and x_2 are states of the same moment continuant *Color* because there is an *moment persistent*[62] C such that $C(w_1) = x_1$ and $C(w_2) = x_2$.

In this case, C is exactly what persists in (4) and (5) while maintaining its identity, and which is capable of changes in a genuine sense.

Finally, besides the concrete urelements that constitute our ontology, we explicitly take into account the conceptual measurement structures in which particular qualities are perceived (and conceived). By employing the theory of conceptual spaces, we can provide a theoretical foundation for the conceptual modeling notion of *attribute values* and *attribute value domains*. Traditionally, in conceptual modeling, value domains are taken for granted. In general, they are considered by taking primitive datatypes as representing familiar mathematical sets (e.g., natural, integer, real, Boolean) and the focus has been almost uniquely on mathematical specification techniques. Once more, we defend that understanding what should be represented is prior to delving on specific ways of specifying it. Whatever constraints should be specified for a datatype must reflect the geometry and topology of the *quality structure* underlying this datatype. Moreover, as discussed in section 6.3.3, by understanding the structure of a certain quality domain, we can derive meta-properties (e.g., reflexivity, asymmetry, transitivity) for the comparative formal relations that are based on this domain.

We emphasize that there are other benefits for conceptual modeling in adopting a notion such as the one of conceptual spaces. For example, in (Gerlst & Pribbenow, 1995), besides their classification of parts as *quantities*, *elements* or *components* (see discussion on section 5.5), the authors

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[62] In the case that a substance bears only one moment of each kind, C can be logically constructed by taking in each world the unique individual x_i that instantiates the super-determinable universal Color. We aimed at generality here to account for situations in which this is not the case, such as in the case of qua individuals.

propose an orthogonal criterion of classification, which is independent of the inherent compositional structure of the entity considered as a whole. The authors name a *Portion* as an aggregate constructed by selecting certain parts of an integral object, according to certain internal properties of this object. For example,

a. the red parts of a painting;
b. the exciting parts of a story;
c. the reddish parts of an object;
d. the dark (colored) parts of a picture;
e. the tall animals in the group.

The resulting *portions* are those parts of the whole that provide the requested value of the relevant property. However, some sentences like (c) and (d) make it clear that the construction of portions with respect to a specific property must be rely on the underlying structure in which a certain property takes it values. Sentence (e) actually exemplifies a situation where the use of *contrast classes* in conceptual spaces can allow for interesting cases of non-monotomic reasoning (Gardenfors 2000; 2004). Since, for instance, although every squirrel is an animal, a tall squirrel is not a tall animal.

By explicitly representing conceptual spaces and their constituent domains and dimensions, one can analyze the same object in different measurement structures. In particular, one can define transformations and projections between conceptual spaces that facilitate knowledge sharing and semantic interoperability. Moreover, one can allow for *context-aware reasoning*, by providing context matrices that emphasize certain dimensions in detriment of others in certain circumstances. In (Raubal, 2004), the author demonstrates how conceptual spaces have been used to model façades of buildings as landmarks for a wayfinding service in Vienna. First, different conceptual spaces are associated with the same landmarks for different purposes. For example, in a database system that stores and manipulates information about the landmarks, the color domain assumed is the RGB model, and a *cultural importance* dimension of the landmarks is considered. For the wayfinding service, a color model based on human perception (HSB, see figures 6.7 and 6.8) is used instead, and the cultural importance dimension is not taken into account. Finally, *weight matrices* are used to highlight different dimensions in different contexts. As discussed by Raubal, people select different landmarks for wayfinding during the day and at night. Thus, while in a *day context*, the best landmark to be used by the service could be a façade with the most contrasting color, in a night context, it can be simply the highest or widest landmark.